

# IAL Biology Unit 6 (WBI16/01) — Complete A\* Study Note

## Practical Skills in Biology II

Built entirely from analysis of 19 past papers and their mark schemes, October 2019 → June 2026.

Everything below is traceable to what examiners actually wrote. Where a mark scheme states an **ACCEPT**, **IGNORE**, **REJECT** or **DO NOT ACCEPT** ruling, it is quoted — in a practical paper those rulings decide almost every mark. Questions are referenced as *series + question number* (e.g. *Jan 2023 Q4(b)*).

### The papers used:

- **Oct 2019** — legacy WBI06 (3 questions, 1 h 30 min), used where the skills are the same.
- **Oct 2020 → Jan 2026** — 17 papers on the current WBI16 specification, each with its mark scheme.
- **June 2026** — the two files supplied do not match: the question paper is **WBI16/01** (flour beetles, forest diversity, training and oxygen, antibiotics), but the mark scheme is for **WBI16/01A**, a different paper (pondweed and Q<sub>10</sub>, height and tidal volume, anaerobic yeast). Both are used. Where this note gives answers to the WBI16/01 paper, they are marked "**model answer — no mark scheme**".

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## 0. THE PAPER, AND WHAT THE MARK SCHEMES REWARD

|                                                          |                                                                                                                                                                                                          |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Code                                                     | WBI16/01                                                                                                                                                                                                 |
| Time                                                     | <b>1 hour 20 minutes</b> (legacy WBI06: 1 h 30 min)                                                                                                                                                      |
| Marks                                                    | <b>50</b>                                                                                                                                                                                                |
| Structure (Jan 2021 onwards)                             | <b>4 questions:</b> two short <b>practical-method</b> questions (Q1, Q2 — about 10 marks each), one <b>data-analysis</b> question (Q3 — about 14 marks), one <b>planning</b> question (Q4 — 15–18 marks) |
| Structure (Oct 2019, Oct 2020, June 2021, June 2026 01A) | <b>3 questions:</b> method (~12–14), data analysis (~15–17), planning (~19–22)                                                                                                                           |
| Maths                                                    | every paper: means, a <b>statistical test</b> (t-test, Spearman, $\chi^2$ , Mann–Whitney), and usually a rate / percentage / magnification                                                               |
| What it examines                                         | the <b>core practicals</b> of Units 4 and 5 <b>and</b> the AS practicals (vitamin C, water potential, tensile strength, mitotic index) — plus a few knowledge marks on the underlying biology            |

**Pace:** 50 marks in 80 minutes = **1.6 min per mark**. You have more time per mark than in Units 4 and 5 — use it to write **every** method point.

### How the 50 marks are distributed, counted across the WBI16 papers

| Question type                                                                        | Typical marks        | Appears in                |
|--------------------------------------------------------------------------------------|----------------------|---------------------------|
| <b>Planning question</b> (preliminary work, method, recording/analysis, limitations) | <b>15–22</b>         | <b>every paper</b>        |
| <b>"Describe an experiment"</b> on a core practical                                  | <b>4–6</b>           | <b>every paper</b>        |
| <b>Null hypothesis + table + graph</b>                                               | <b>7 (1 + 3 + 3)</b> | <b>every paper</b>        |
| <b>Statistical test calculation + conclusion</b>                                     | <b>4–5</b>           | <b>every paper</b>        |
| <b>Abiotic / biotic variable + control + effect</b>                                  | <b>3–5</b>           | <b>every paper</b>        |
| <b>Theory mark(s)</b> on the biology behind the practical                            | <b>2–4</b>           | <b>every paper</b>        |
| <b>Risk and how to reduce it / ethics</b>                                            | <b>1–2</b>           | <b>nearly every paper</b> |
| <b>Extend / improve / why not valid</b>                                              | <b>2–4</b>           | <b>every paper</b>        |

### Statistical test used in the data question:

| Test                                                                     | Papers                                                                                                                                |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Student's t-test</b>                                                  | Oct 2020, Jan 2021, Jan 2022, June 2022, Jan 2023, June 2023, Oct 2023, Jan 2024, Oct 2024, Oct 2025, Jan 2026, June 2026 — <b>12</b> |
| <b>Spearman's rank correlation</b>                                       | Oct 2021, Oct 2022, Jan 2025, June 2025, June 2026 (01A) — <b>5</b>                                                                   |
| <b>Chi-squared</b>                                                       | June 2022, June 2024, Jan 2025 — <b>3</b>                                                                                             |
| <b>Mann–Whitney U</b>                                                    | Oct 2019, June 2021 — <b>2</b>                                                                                                        |
| Other (odds ratio, SD, diversity index, growth-rate constant, $Q_{10}$ ) | June 2023, June 2025, June 2026, Jan 2023                                                                                             |

**The t-test is the default.** Expect it; practise it until it takes four minutes.

### The five things the Unit 6 mark schemes reward that students miss

1. **Quantities, not "amount".** "*Ignore amount*", "*Ignore amount unqualified*" appear in scheme after scheme. Write **volume, mass, concentration, number, distance, time** — with a unit.
2. **A stated number.** The schemes want **five** concentrations / temperatures, a **stated** time (e.g. 24–72 h), a **stated** temperature range (e.g. 5–50 °C). "Different temperatures" does not score; "5, 15, 25, 35 and 45 °C" does.
3. **How, not just what.** A controlled variable scores only when you say **how** it is controlled: "*variable with suitable control method described*" — temperature → **thermostatic water bath**; pH → **buffer**; light → **same wattage bulb at a fixed distance**; mass → **balance**.
4. **Directional effects.** For "the effect if not controlled", the scheme demands "*description of expected effect on the dependent variable ... must be directional*" (Jan 2021, June 2021, Jan 2024). "It would affect the results" = 0. "A higher temperature would increase enzyme activity, so more CO<sub>2</sub> would be produced" = 1.
5. **Graph and test must match the design.** "*bar graph + t-test for difference*" when two groups are compared; "*line / scatter graph + correlation test*" when there are several values of a continuous IV (Oct 2023, Oct 2024, June 2025, Jan 2026). A mismatch costs the mark.

# PART A — THE PLANNING QUESTION

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## 1. The anatomy of the planning question

The last question always gives you a **context**, a **student's observation** and a **hypothesis**, then asks you to "*plan an investigation to find evidence to support or reject this hypothesis*" under fixed headings:

| Part | What it asks                                                                                                                                                            | Marks       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| (a)  | <b>Preliminary practical work</b> "to ensure your method would provide quantitative / meaningful results" — or, in some papers, a <b>safety issue / ethical concern</b> | 2–3         |
| (b)  | <b>A detailed method</b> , "including how you would control and monitor important variables"                                                                            | <b>8–10</b> |
| (c)  | <b>How results are recorded, presented and analysed</b>                                                                                                                 | 3–4         |
| (d)  | <b>Limitations</b> of your method                                                                                                                                       | 2–4         |

The method part is marked as "**an answer that includes eight / nine / ten of the following**" — a list of about 10–12 creditable points. It is **not** levels-marked (except Oct 2019, which added 2 QWC marks). So every separate, correct point you write is worth a mark, up to the maximum.

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## 2. Part (a): preliminary work — the "find a suitable..." rule

The schemes are explicit: "*Ignore practise the method*" (Oct 2020, June 2022); "*Ignore stated methods – credit is for finding out*" (June 2021); "*Do not award method statements*" (Oct 2022). A preliminary point is always a "**find a suitable ...**" statement.

**The template (2–3 marks):**

**Find a suitable** {range / concentration / mass / number / age} of {the IV / the organism} **to give measurable results** · **find a suitable** {temperature / pH / time} for {respiration / germination / growth / enzyme activity} · **find a suitable method to measure** the dependent variable

**The June 2026 (01A) rule — say what the "suitable" is for.** "*If there is no mention of what the suitable conditions are for, then can give only two marks if there are three ideas, one mark if two, none if one.*" Always add "**... for measurable respiration to occur**" (or whatever the process is).

**Preliminary points credited in the papers:**

| Plan                                              | Preliminary points in the scheme                                                                                                                                                                               |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cabomba — light and photosynthesis (Oct 2020)     | a method for <b>measuring oxygen production</b> (" <i>IGNORE counting oxygen bubbles</i> ") · a <b>suitable time period</b> (10 min – 8 h) · a method for <b>varying light intensity</b>                       |
| Wheat — ABA and amylase (Jan 2021)                | suitable <b>ABA concentration range</b> · method for <b>measuring starch digestion</b> · suitable <b>temperature / pH</b> · suitable <b>timescale</b>                                                          |
| Aplysia — gill withdrawal (June 2021)             | a method of <b>touching the gills</b> · <b>conditions in which the slugs are active</b> · a method to <b>decide when the gills have re-emerged</b>                                                             |
| Pineapple juice and <i>E. coli</i> (Oct 2021)     | suitable <b>concentration / volume of juice</b> · <b>timescale for growth</b> · <b>temperature / pH</b> · a way to <b>count bacteria</b>                                                                       |
| Yeast — temperature and respiration (Jan 2022)    | suitable <b>mass / concentration of yeast</b> · a <b>method for measuring CO<sub>2</sub> / O<sub>2</sub></b> · a <b>redox indicator</b> · a <b>range of temperatures</b>                                       |
| Mung bean RQ (June 2022)                          | suitable <b>ages of seedlings</b> · <b>temperature</b> · <b>time to measure gas</b> · <b>mass / number of seedlings</b>                                                                                        |
| Polychaete habituation (Oct 2022)                 | <b>time for the animal to start feeding</b> · <b>conditions for feeding</b> · a method for applying a <b>touch of suitable force</b> · a method of <b>measuring fan withdrawal</b>                             |
| Grasshopper populations (Jan 2023)                | a way to <b>identify the species</b> · a way to <b>make them move</b> (camouflage) · the <b>time of day / month</b> they are active · a <b>sampling / counting method</b> · suitable <b>weather</b>            |
| Antibacterial leaves (June 2023)                  | suitable <b>mass / age of leaves, extract concentration, extraction method</b> · suitable <b>temperature / pH / medium / time / species of bacteria</b> · a method to <b>measure inhibition</b>                |
| Cabomba light (Oct 2023), Elodea depth (Oct 2024) | suitable <b>mass / length of plant</b> · a way to <b>change light intensity / depth</b> · suitable <b>temperature / time</b> · a method to <b>measure gas volume</b>                                           |
| Watermelon RQ (Jan 2024)                          | a way of <b>germinating seeds / checking viability</b> · suitable <b>temperature</b> · <b>mass of seeds</b> for a measurable volume · a <b>CO<sub>2</sub> absorber</b> · a method to <b>measure gas volume</b> |
| Mung bean vs pea respiration (June 2024)          | <b>pre-treatment</b> (soaking, rinsing) · conditions for <b>germination</b> · <b>time</b> for seeds to start respiring · <b>timescale</b> for bubble movement · <b>number / mass</b> of seeds                  |
| Blowfly larvae (Jan 2025)                         | suitable <b>mass / age / species of larvae</b> · <b>temperature</b> · a method to <b>measure oxygen consumption</b> · <b>timescale</b>                                                                         |
| Wheat varieties and starch digestion (Oct 2025)   | a method to <b>measure starch digestion</b> · <b>conditions</b> (temperature / pH) · <b>time</b> · <b>mass of wheat</b> / starch concentration                                                                 |

| Plan                             | Preliminary points in the scheme                                                                                                                                                             |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anaerobic yeast (June 2026, 01A) | suitable <b>strain, concentration / mass of yeast, mass of sugar, timescale, temperature / pH</b> — each "for measurable respiration to occur" — <i>"IGNORE redox indicator suitability"</i> |

### 3. Part (b): the method — the universal mark points

Across all 18 planning schemes, the credited points fall into the same families. Write one sentence for each:

| #    | Mark point                                                        | How the scheme phrases it                                                                                                                                                    |
|------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Dependent variable, clearly stated with how it is measured</b> | <i>"clear statement of the dependent variable, e.g. volume of O<sub>2</sub> released per unit time" — "Ignore amount", "Ignore rate" on its own</i>                          |
| 2    | <b>Independent variable with a range</b>                          | <i>"use of five different concentrations", "at least two temperatures stated (20–35 °C)", "use both types of seed", "one group with and one without a fitness programme"</i> |
| 3    | <b>How the IV is changed</b>                                      | <i>"description of method to provide 5 different intensities of light" — moving lamp / bulb wattage / filters</i>                                                            |
| 4    | <b>Apparatus / technique for measuring the DV</b>                 | photosynthometer, respirometer (labelled diagram accepted), gas syringe, colorimeter, stage micrometer, starch agar + iodine, quadrats                                       |
| 5    | <b>Standardise the organisms / material</b>                       | <i>"same mass / length of pondweed", "same mass of seeds", "same strain / source of yeast", "maggots hatched at the same time"</i>                                           |
| 6    | <b>Acclimatise / equilibrate</b> before measuring                 | <i>"plant given time to acclimatise", "seeds left to equilibrate", "incubate yeast and indicator separately to reach same temperature"</i>                                   |
| 7    | <b>Stated time</b> for the measurement                            | <i>"volume of gas collected in stated time (3–30 min)", "incubate 24–72 hours"</i>                                                                                           |
| 8    | <b>Calculation of rate</b>                                        | <i>"distance ÷ time", "volume ÷ time", "1 ÷ time for colour change", "change in length ÷ time", "<math>\pi r^2 d \div t</math>"</i>                                          |
| 9–10 | <b>Two controlled variables</b>                                   | <i>"two variables that need to be controlled (1)"</i>                                                                                                                        |

| #     | Mark point                             | How the scheme phrases it                                                                                                                     |
|-------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 10–11 | <b>... and how each is controlled</b>  | <i>"description of how these variables are controlled"</i> (see §22)                                                                          |
| 12    | <b>Control / comparison</b>            | <i>"use of a control for comparison", "with and without pineapple juice — accept heated juice as a control", "with and without soda lime"</i> |
| 13    | <b>Aseptic technique</b> (any microbe) | <i>"flaming neck of bottles / sterilise loop / work near a Bunsen flame" — "ignore taping dishes"</i>                                         |
| 14    | <b>Repeats → mean AND SD</b>           | <i>"repeat the whole experiment to calculate mean and SD / error bars" — from 2023 the scheme often wants <b>SD</b> as well as the mean</i>   |
| 15    | <b>Ethics / welfare</b> (animals)      | <i>"the method of touching the gills must not cause pain / damage"</i>                                                                        |

#### Two structural rules:

- *"Do not piece together the dependent variable"* (Jan 2024) — state the DV **in one clear sentence**, with what is measured and per what time.
- Writing the RQ formula, a named test, or a named indicator often carries its **own** mark — name things.

## 4. Part (c): recording, presenting, analysing — the 3-mark formula

Every scheme from 2021 uses the same three points:

1. **A table for raw data with headings and units, and means calculated from repeats** — *"Do not accept units in the body of the table"; "Reject rate in heading" when the table is for raw data (Oct 2024); "Do not accept a calculated rate / recovery rate heading in table (not raw data)" (June 2025).*
2. **A graph of the right type with labelled axes.**
3. **A named statistical test of the right type.**

The pairing rule, stated in the schemes (Oct 2023, Oct 2024, June 2025, Jan 2026):

| Your design                                                                                             | Graph            | Test                                                             |
|---------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------|
| <b>Two groups / two treatments</b> compared (e.g. two varieties, with vs without, trained vs untrained) | <b>bar graph</b> | <b>t-test</b> (or Mann–Whitney U) — <i>"test for difference"</i> |

| Your design                                                                               | Graph                        | Test                                                   |
|-------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------|
| <b>Several values of a continuous IV</b> (e.g. 5 light intensities, 5 temperatures, ages) | <b>line or scatter graph</b> | <b>correlation test — Spearman's rank</b> (or Pearson) |
| <b>Counts in categories</b> (choice chamber, T-maze, presence / absence)                  | bar chart                    | <b>chi-squared</b>                                     |

Examples: RQ at several ages → line graph + correlation; "*t test if only two ages*" (June 2022). "*RQ has no unit.*"

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## 5. Part (d): limitations — the bank

The schemes credit limitations of **three kinds**. Give one of each.

### (i) Measuring the DV is hard

- "*difficult to measure (small) volumes / distances of gas with precision*" — photosynthesis and respiration plans every time
- "*oxygen dissolves in water / some  $O_2$  is used in respiration / some gas may be  $CO_2$* " (photosynthesis)
- "*difficult to judge the end point / colour change*" (redox indicators, DCPIP, iodine, colour standards)
- "*difficult to measure small clear areas / clear zone may not be circular*" (starch agar, zones of inhibition)
- "*difficult to determine full re-emergence / extent of withdrawal*", "*difficult to standardise the touch*" (habituation)
- "*difficult to measure a curved shoot / judge where the shoot starts*" (growth)
- "*difficult to read the trace / leaking air / soda lime stops working*" (spirometer)

### (ii) A variable is hard to control

- "*difficult to control temperature*" — **germinating seeds and respiring larvae produce heat**; lamps heat water
- "*difficult to control surface area of leaves*"
- "*difficult to control age / viability / genetic variation*" of seeds, larvae, animals
- "*the other aspects of volunteers' lives*" — diet, stress, health, previous fitness (June 2025)

### (iii) The biology may change during the experiment

- "*seeds / yeast may change from aerobic to anaerobic respiration*"
- "*build-up of waste products / ethanol may slow respiration*"

- "gases expand at higher temperatures", "CO<sub>2</sub> is soluble so gas volume is underestimated" (Jan 2022)
- "contamination / hard to maintain aseptic conditions"
- "only tested against one species of bacteria", "bacteria cultured aerobically but the gut is anaerobic / not at body temperature" (June 2023)
- "conditions in the river change during the day, so the lab results may not represent the field" (Oct 2023)
- "if mung beans germinate faster than peas, the peak respiration may be at different times after soaking" (June 2024)

"Ignore bubbles / escape of gas / errors in measuring" unqualified (Oct 2023) — every limitation must say **what** is hard and **why** it matters.

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## 6. The planning questions, one by one

### 6.1 Photosynthesis plans

**Cabomba** — light intensity (Oct 2020, 10 marks; Oct 2023, 9 marks); **Elodea** — depth (Oct 2024, 9 marks)

DV: **volume of O<sub>2</sub> / gas released per unit time** ("IGNORE counting bubbles"; "Do not allow carbon dioxide") · **photosynthometer / gas syringe / capillary** — labelled diagram accepted · **same mass / length** of plant · **five light intensities** — **moving the lamp / bulb wattage**; for depth, **tubes of water of different depths** or equivalent light intensities · **sodium hydrogencarbonate** so CO<sub>2</sub> is in excess · **acclimatise** · **stated collection time** (5 min – 24 h) · control **temperature** — **thermostatic water bath / heat shield / LED bulb** ("Ignore AC room", Oct 2023), **pH** — **buffer**, **wavelength** — **filter** · **repeats** → **mean and SD** · **rate = volume ÷ time** ( $\pi r^2 d \div t$ )

### 6.2 Respiration plans

**Yeast** — temperature (Jan 2022, 8 marks)

DV: **volume of CO<sub>2</sub> per unit time / O<sub>2</sub> used per unit time / time for redox indicator to change colour** · apparatus (gas collection / respirometer with soda lime / **tubes in a water bath before mixing**) · **control mass / concentration of yeast** · incubate for a set time and record gas volume or time to colour change · **five temperatures in 5–55 °C** · two controlled variables (**pH** — **buffer**, **glucose concentration** — **balance**, **volume of indicator**, **strain of yeast**) · **repeats** · **rate = 1 ÷ time, or volume ÷ time**

**Mung bean RQ with age** (June 2022, 8 marks); **watermelon diploid vs tetraploid RQ** (Jan 2024, 9 marks)

DV **volume of gas / distance moved by meniscus** · apparatus (**respirometer / test tube with capillary / gas syringe**) · seedlings of **different ages** (or both seed types) · **same mass of seeds** · record distance in a set time · **run with and without soda lime** · **find  $O_2$  used = distance with soda lime  $\times \pi r^2$ ;  $CO_2$  produced =  $O_2$  used – net change without soda lime** · temperature controlled with a **thermostatic device** · **surface-sterilise seeds** (bleach) · **acclimatise** ·  **$RQ = CO_2 \text{ produced} \div O_2 \text{ consumed}$**

**Mung beans vs peas (June 2024, 8 marks):** DV **distance moved by the bubble / volume of  $O_2$  used in a given time** · **seed preparation** (soaking, rinsing, surface sterilisation) ·  **$CO_2$  absorber** ("*Not sodium hydrogen carbonate*") · **equilibrate** · rate calculated · one **abiotic** variable controlled and how ("*Ignore pH*") · one **biotic** variable (age, mass, variety) · **use both seed types**.

**Blowfly larvae on liver vs muscle (Jan 2025, 9 marks):** DV **oxygen consumption per unit time** · **respirometer** · **same mass of larvae** ("*Ignore number of larvae / weight*") · **soda lime** ("*reject sodium hydrogen carbonate*") · capillary / syringe detail · **rate per unit mass** · **repeat with the other tissue** · two variables + control (temperature, age, species, mass of meat, cube size) · **repeats** → **mean and SD / error bars**.

**Anaerobic yeast (June 2026, 01A, 9 marks):** DV ( $CO_2$  volume / ethanol / time for indicator) · **anaerobic conditions** — **layer of oil, airlock, sealed apparatus** · **mix yeast and sugar** · **equilibrate** · **same stated temperature (20–35 °C)** · same **volume** of sugar and yeast · same **sugar concentration** · same **strain / source** of yeast — "*DO NOT ACCEPT species or age*" · method of measuring DV · **repeats** → **mean and SD**.

### 6.3 Microbiology plans

**Pineapple juice and *E. coli* growth (Oct 2021, 8 marks)**

DV **number of bacteria / turbidity / absorbance** with time · **making the juice** · **broth with and without juice** — "*accept heated juice / water as a control*" · **aseptic technique** (flame the neck of bottles, work near a Bunsen) · **samples at suitable intervals** · counting — **colorimeter / serial dilution and plating** · **growth rate calculated** · two variables + control (temperature, pH, pineapple variety/age, juice concentration, culture volume/age, medium) · **repeats**

**Antibacterial leaves — *Piper betle* (June 2023, 9 marks)**

DV **zone of inhibition** · **preparing the extract** · **bacterial lawn / broth / pour plate** · **applying the extract** — **wells / soaked discs** · measuring the zone — **ruler, several diameters / grid** · **aseptic technique** ("*ignore taping dishes*") · **control** (disc with solvent only) · **incubate at a stated temperature** — **not above 30 °C** · two variables + control (incubation time, disc size, medium, bacterial species, mass of leaves) · **repeats** → **mean, SD / error bars**

**Ampicillin vs erythromycin (June 2026, 01, 9 marks) — model answer, no mark scheme.** Use the same template as *Piper betle*: lawn of one *E. coli* strain on nutrient agar, **discs of the same**

**diameter** soaked in **equal concentrations** of each antibiotic, a **sterile-water control disc**, aseptic technique, incubate at 25 °C for 24–48 h, measure **zone diameter / area**, repeat to obtain means and SD, bar graph + t-test. (To study "decrease growth faster", sample a broth culture with each antibiotic at intervals and measure **turbidity with a colorimeter**.)

## 6.4 Seeds, enzymes and plant growth

### ABA and amylase production in wheat (Jan 2021, 8 marks)

DV **size of the clear area** (or absorbance with a colorimeter) · **aseptic technique** — **rinse endosperm in sodium hypochlorite / sterile water** · **soak half-grains in ABA**, place on **starch agar** · **incubate 24–72 h** · measure clear area — **trace it or take several diameters** · two variables + control (pH, temperature, humidity, light, variety, age of seed, mass) · **repeats** · **five ABA concentrations**

### Two wheat varieties digesting starch (Oct 2025, 9 marks) — two accepted routes:

- *Starch agar route*: DV **area / diameter of clear zone per unit time** ("*Do not accept ZOI*" — it is a **clear zone**, not a zone of inhibition) · **starch agar** · **germinate grains / soak in gibberellin** · **standardised preparation** (surface-sterilise, cut, cut side down) · **aseptic technique** · **iodine solution** to reveal the clear zone ·  $\text{rate} = \text{area} \div \text{time}$  · two variables + control · **both varieties** · **repeats** → **mean and SD**.
- *Extract route*: DV **proportion of starch remaining after a set time** · grind / filter to form an extract · **iodine + colour standards / colorimeter and calibration curve** · rate · controls · both varieties · repeats.

**Drumstick seedlings and temperature (Jan 2026, 9 marks)**: DV **shoot length after a stated time** · **same number, age and source of seeds** · **at least two temperatures in 20–35 °C** · **standardised watering** (same volume, same timing) · **measuring the shoot** ("*Ignore root / leaf / mass*") · rate of growth · two variables + control ("*Ignore pH and buffer*"; "*Accept carbon dioxide in excess*") · **repeats** → **mean and SD**. Ethics (4(a)): **not an endangered species / a farmed food crop** — "*Ignore ref to pain / nervous system.*"

## 6.5 Behaviour plans (habituation)

### Aplysia gill withdrawal (June 2021, 10 marks); polychaete fan withdrawal (Oct 2022, 8 marks)

DV **time for re-emergence / extent of withdrawal** · **different frequencies / numbers of touches** · **same intensity of touch** — **same person, same glass rod / cotton bud, same force** · **how to judge full re-emergence** (sketch the extended gills first) · **the touch must not cause pain or damage** · **acclimatise** before and **between** touches · supply **organic particles** so the animal feeds (polychaete) · two variables + control (**water temperature, pH, salinity, depth, light; age, sex, species**) · **repeat with different animals**

Limitations: **standardising the touch, judging full re-emergence, age / surface area of gills, other stimuli — sound, shadows, vibration, water currents, animals already habituated.**

## 6.6 Field and population plans

### Clover flowers and soil water (Oct 2019, 10 + QWC)

DV **number of flowers per plant** · **transect / random quadrats on ridges and furrows** · method of counting **mean flowers per plant** · **standardised soil sampling** (same depth / position) · **soil water — moisture meter or dry and reweigh** · two variables (light, temperature, pH, minerals, wind, grazing, competition, species) **and how monitored** · **sample within a short period (same day)** · **repeat in other fields**

### Grasshopper populations on two gravel sites (Jan 2023, 8 marks)

DV **number per unit area** · **standardised sample sites of known area** on both sites · **standardised counting method and time** (nets, quadrats, camera, **mark–release–recapture / Lincoln index**) · **monitor two variables** (temperature, light, humidity, rainfall, time of day, slope / aspect) and describe how · **repeats** at each site and on **different days / months (November–March)** · **calculate population density**

Limitations: **identifying the species, making camouflaged insects move, counting the same individual twice, endangered — can only be sampled once / population changes.**

## 6.7 Human physiology plan

### Fitness programme and recovery (June 2025, 9 marks + 2 ethics/safety)

(a) **informed consent** / students cannot be forced · **sterilise the mouthpiece** / check health / change the CO<sub>2</sub> absorber (b) **one group with and one without the programme** · DV **time for the trace (breathing rate, tidal volume) to return to resting** · apparatus detail (**nose clip**, calibrate, practise on the apparatus) · **soda lime in the canister** · **sterilise / replace mouthpiece** for each student · **find resting breathing rate / tidal volume first** · **standardised exercise** (e.g. one minute of step-ups) · measure time to return to rest · two variables + control (**BMI, sex, height, mass, age, smoker, room temperature, humidity**) · **repeats** → **mean and SD**

## 7. Model plan skeleton — use for any hypothesis

- (a) Preliminary: find a suitable [range of IV] and a suitable [time / temperature] for measurable [process]; find a suitable method to measure [DV].
- (b) Method:
1. DV = [quantity] per unit time, measured with [apparatus].
  2. Vary IV: [5 stated values], produced by [method]. (Or: two groups X and Y.)
  3. Standardise the organism: same [mass / age / length / strain / source].
  4. Acclimatise / equilibrate for [10 min] at the test conditions.
  5. Measure for a stated time [e.g. 5 min]; rate = [quantity] ÷ time.
  6. Control: [variable 1] by [method]; [variable 2] by [method].
  7. Control group: [without the treatment / heated extract / water].
  8. Aseptic technique OR welfare precaution, as appropriate.
  9. Repeat each [value] 3–5 times; calculate mean and standard deviation.
- (c) Table of raw data (headings + units, no units in the body) with means; [bar graph + t-test] OR [line/scatter graph + Spearman's rank].
- (d) Limitations: one measurement difficulty; one uncontrollable variable; one change during the experiment.
-

## PART B — "DESCRIBE AN EXPERIMENT" AND THE CORE PRACTICALS

### 8. The "describe an experiment" template

Q1 and Q2 almost always open with *"Describe how [the student] could carry out an investigation into..."* (4–6 marks). It is marked as **"a description that includes five / six of the following"**. The mark points are close to identical across every practical:

| # | Mark point (as worded in the schemes)                                   | What you write                                                                                                                                                                                                            |
|---|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>"clear statement of the dependent variable"</b>                      | "the dependent variable is the <b>volume of oxygen</b> collected <b>per minute</b> " — a measurable quantity, never "rate of photosynthesis" ( <i>"Ignore rate" — Jan 2023; "IGNORE rate of respiration" — Oct 2019</i> ) |
| 2 | <b>"use of five different / stated {concentrations / temperatures}"</b> | name them: "0, 0.5, 1.0, 1.5 and 2.0 mol dm <sup>-3</sup> "; include <b>zero</b> as one of them where it makes sense ( <i>"Accept use of zero ... as one of the concentrations"</i> )                                     |
| 3 | <b>suitable apparatus</b> for the organism / measurement                | photosynthometer, respirometer, starch-agar plate, colorimeter, spirometer, Petri dish...                                                                                                                                 |
| 4 | <b>"{same / stated} time"</b>                                           | e.g. "after <b>24 hours</b> "; the schemes give a window (e.g. 3–30 min for gas; 24–72 h for bacteria; 1 h–3 days for brine shrimp)                                                                                       |
| 5 | <b>"identification and control of one variable"</b>                     | variable <b>plus</b> method: temperature → <b>thermostatic water bath</b> ; pH → <b>buffer</b> ; light → <b>bulb at fixed distance / same wattage</b>                                                                     |
| 6 | <b>a second controlled variable</b> (on 6-mark versions)                | e.g. "same <b>mass</b> of yeast measured on a <b>balance</b> "                                                                                                                                                            |
| 7 | <b>"repeats (for each treatment) and calculate a mean"</b>              | <i>"Accept repeats to calculate SD", "Accept average"</i>                                                                                                                                                                 |
| 8 | <b>"method of calculating rate"</b> where appropriate                   | "rate = volume ÷ time", "rate of growth = change in length ÷ time"                                                                                                                                                        |
| 9 | <b>acclimatise / equilibrate</b>                                        | "leave the pondweed for <b>5 minutes</b> to acclimatise", "incubate yeast and indicator <b>separately</b> to reach the same temperature"                                                                                  |

**Rule of thumb:** a 6-mark describe question has about 7–8 creditable points; write **all eight**. Seven short numbered sentences beat one long paragraph.

**The five words that lose marks here:** *amount* ("ignore amount unqualified"), *size* ("IGNORE size"), *suitable* without a value ("do not accept suitable"), *rate* as the DV, and *different* without numbers.

## 9. Photosynthesis — pondweed (*Elodea*, *Cabomba*)

**Tested:** Oct 2020 plan (light), Oct 2021 Q1 (wavelength), Oct 2022 Q1 (temperature), Oct 2024 plan (depth), Oct 2023 plan, June 2026 01A Q1 (temperature + Q<sub>10</sub>).

**Mark points (merged from Oct 2021 Q1(a), Oct 2022 Q1(a), June 2026 01A Q1)**

1. **Pondweed set up with suitable apparatus to collect / measure the volume of gas** — "Accept name of apparatus eg photosynthometer / diagram... funnel + tube + (gas) syringe".
2. **IV stated in values** — five temperatures **within 5–50 °C** (Oct 2022); **at least two coloured filters, or one filter and white light** / named wavelengths **480–750 nm** (Oct 2021).
3. **Sodium hydrogen carbonate added (in excess)** — "Accept ... bicarbonate / potassium hydrogen carbonate". Its reason: "so (concentration of) CO<sub>2</sub> is not limiting" — "DO NOT ACCEPT enough CO<sub>2</sub>" (June 2026 01A).
4. **Time for pondweed to acclimatise** — reason: "so enzymes are working at the desired temperature".
5. **Volume of gas collected in {same / stated} time** — e.g. 3–30 min.
6. **Control of one variable** — pH – buffer; mass or length of pondweed – measure; light intensity – bulb at set distance; heat shield; wavelength – filter.
7. **Method of calculating rate** — volume ÷ time.
8. **Repeats and means** — "so that mean AND standard deviation can be calculated" (June 2026 01A (a)(iv)).

### Bubble counting vs volume

The June 2026 01A student counted **bubbles per minute**. The improvement mark: "(measuring / calculating) volume of oxygen / gas ... use of (gas) syringe or movement of bubble in capillary tubing". Bubbles vary in size, so volume is more valid. In a plan, "IGNORE counting bubbles" for the DV mark.

**Explaining controls — the "variable + why" pairs (June 2026 01A (a)(iii), 4 marks from two explanations)**

| Control                                          | Why (the second mark)                                       |
|--------------------------------------------------|-------------------------------------------------------------|
| pieces of equal <b>length / mass</b>             | <b>so the number of cells photosynthesising is the same</b> |
| <b>measured distance</b> between lamp and beaker | <b>so light intensity is equal</b>                          |
| add <b>sodium hydrogen carbonate</b>             | <b>so CO<sub>2</sub> is not limiting / is in excess</b>     |
| leave to <b>acclimatise</b>                      | <b>so enzymes are working at the desired temperature</b>    |

**Theory marks attached**

- **Temperature** (June 2026 01A (b)(i), 4): rate increases up to a point; photosynthesis relies on **enzymes, e.g. RUBISCO**; more **kinetic energy**; **more collisions per unit time** / more ES complexes; at high temperature enzymes **denature**, active site changes shape.
- **Q<sub>10</sub>** (01A (b)(ii)): rate at 30 °C = 17.6 × Q<sub>10</sub>; with Q<sub>10</sub> = 2 → **35.2 bubbles per minute** ("ACCEPT any value between 35.2 and 52.8").
- **Grana** (Oct 2021, 3): stacks of **membrane** → **large surface area** for photosystems; contain **pigments** to absorb light; contain **electron carriers / enzymes** for the light-dependent reactions.
- **Oxygen production** (Oct 2022, 3): **photolysis**; water split:  $\text{H}_2\text{O} \rightarrow 2\text{H}^+ + 2\text{e}^- + \frac{1}{2}\text{O}_2$ ; **electrons removed from water**; two oxygen atoms combine to form O<sub>2</sub>. "Ignore light dependent reaction unqualified."

**Aquatic plant temperature control:** use a **thermostatic water bath** or a **beaker of water with a thermometer, topping up with hot/cold water**. For aquatic plants the schemes "Ignore AC room" — the water temperature is what matters. A **heat shield** (a beaker of water between lamp and plant) stops the lamp heating the water.

## 10. Respiration — the respirometer

**Tested:** Oct 2019 Q1 (mung beans), Jan 2021 Q2 (maggots), Jan 2022 plan (yeast), June 2022 plan (mung RQ), Jan 2024 plan (watermelon RQ), June 2024 plan (mung vs pea), Jan 2025 plan (blowfly larvae), June 2026 01A plan (anaerobic yeast).

**The set-up**

- Organisms (germinating seeds, maggots, woodlice) in a boiling tube on **gauze**, above **soda lime / potassium hydroxide** which **absorbs CO<sub>2</sub>**.

- Tube connected to a **capillary tube (manometer) with coloured fluid**; a **control tube with glass beads of the same mass/volume** (no living organisms).
- Whole thing in a **thermostatic water bath**; **equilibrate** for ~10 min with the screw clip / tap open, then close.
- O<sub>2</sub> used → pressure falls → fluid moves **towards** the organisms. Measure **distance moved in a stated time**.

### Mark points (Oct 2019 Q1(a))

1. **DV identified** — "oxygen consumption / oxygen used" ("*IGNORE rate of respiration*").
2. **Method of measuring DV** — movement of fluid using a respirometer (diagram accepted).
3. **Soda lime / KOH to absorb CO<sub>2</sub>**.
4. **Rate with reference to distance and time**.
5. **At least two temperatures within 5–50 °C**.
6. **Thermostatic water bath** ("*ACCEPT incubator or air conditioned room*").
7. **Repeats and mean**.

### The calculations

**Volume of gas from the distance moved** (Jan 2021 Q2(a)(i)):

volume =  $\pi r^2 \times \text{distance}$ . Mean distance = 4.4 mm,  $r^2 = 0.01 \text{ mm}^2$  → volume per minute = **0.14 mm<sup>3</sup> min<sup>-1</sup>** (0.138 gets 2 marks).

**RQ** (Jan 2021 Q2(a)(ii)):

RQ = CO<sub>2</sub> produced ÷ O<sub>2</sub> used = 0.11 ÷ 0.14 = **0.79** (0.786 / 0.8 accepted).

**RQ with and without soda lime** (the plan version — June 2022, Jan 2024):

Run **with** soda lime → the fluid moves distance **x** = O<sub>2</sub> used. Run **without** soda lime (water / glass beads instead) → the fluid moves **y** = O<sub>2</sub> used – CO<sub>2</sub> produced. So CO<sub>2</sub> produced = **x – y**, and **RQ** = **(x – y) ÷ x**.

**Why glass beads / remove soda lime** (Jan 2021 Q2(a)(iii)): "*soda lime absorbs carbon dioxide so must be removed so carbon dioxide produced can be measured*".

### Theory: the role of oxygen (Oct 2019 Q1(c), 3)

- oxygen required for **aerobic respiration**;
- it is the **final electron acceptor** in **oxidative phosphorylation**;
- at low O<sub>2</sub> **anaerobic respiration** occurs.

## Variables for a respirometer (Jan 2021 Q2(b)(i))

Biotic: **mass of maggots, species, age**. "Ignore size, sex, type, genetic variation, microbes in tube, competition, disease." Note: **two abiotic = 0** when biotic was asked.

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## 11. Respiration — redox indicator (yeast)

**Tested:** June 2024 Q1, June 2025 Q1.

The indicator (**TTC, DCPIP, methylene blue**) accepts the electrons / hydrogen that would normally go to NAD/FAD and **changes colour** when reduced. Faster respiration → faster colour change.

### Mark points (June 2024 Q1(b) and June 2025 Q1(b))

1. **DV = time for the indicator to change colour** ("ignore stated colours"; "Ignore any stated colour change").
2. **Five temperatures, 5–45 °C (2024) / 0–60 °C (2025)**.
3. **Thermostatic water bath** ("Accept waterbath set at a stated temperature").
4. **Incubate yeast and indicator separately to reach the same temperature** — "equilibrate / acclimatise".
5. **Named indicator:** TTC / DCPIP / methylene blue.
6. **Same strain / species / age / mass of yeast / volume of culture** ("Ignore concentration" in 2025).
7. **Add glucose / sucrose / respiratory substrate** ("Ignore nutrients").
8. **Same volume / mass of indicator** ("Ignore amount / number of drops").
9. **Stopwatch / timer** — start when the indicator is added, stop when it has changed colour.
10. **Repeats and means / SDs**.

### Theory marks attached

- **Where** the electrons would normally go (June 2024 (a)): **cristae / inner mitochondrial membrane**; the natural carriers are **NAD / FAD** ("Do not accept NADP"; "No mark if one correct and one incorrect answer").
  - **Temperature effect** (June 2024 (c), 3): more **kinetic energy**; **more collisions** between enzyme and substrate / more ES complexes; above the optimum enzymes **denature**, active site changes shape.
  - **ATP from anaerobic respiration** (June 2025 (c), 3): the co-enzyme is **NAD** ("Not co-enzyme A / FAD"); electrons transferred to the co-enzyme; phosphate added to ADP to form ATP by **substrate-level phosphorylation** — "not oxidative phosphorylation".
-

## 12. The spirometer — breathing and oxygen uptake

Tested: Oct 2021 Q2, Oct 2023 Q2, June 2025 plan (fitness), June 2026 01A Q2.

### Reading the trace

| Quantity              | How to get it                                                                                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Breathing rate        | count peaks in a known time → breaths per minute (Oct 2021: <b>22.5 min<sup>-1</sup></b> , "Accept 21 / 24")                                                      |
| Tidal volume          | height of one peak-to-trough (Oct 2021: 0.6–0.75 dm <sup>3</sup> ; June 2026 01A: 0.45–0.55 dm <sup>3</sup> )                                                     |
| Minute ventilation    | <b>tidal volume × breathing rate</b> (Oct 2021: 13.2–18.0 dm <sup>3</sup> min <sup>-1</sup> ; June 2026 01A: 4.7–6.4)                                             |
| Rate of oxygen uptake | the <b>gradient of the line joining the troughs</b> (or peaks): fall in volume ÷ time (Oct 2023: <b>6.1–6.5 × 10<sup>-3</sup> dm<sup>3</sup> s<sup>-1</sup></b> ) |

**Units ruling** (Oct 2023 Q2(b)): "Not dm<sup>3</sup>/s or dm<sup>3</sup>/min. Allow per second or per minute." Write **dm<sup>3</sup> s<sup>-1</sup>** or **dm<sup>3</sup> per second**. The scheme gives the equivalent ranges in dm<sup>3</sup> min<sup>-1</sup>, cm<sup>3</sup> s<sup>-1</sup> and cm<sup>3</sup> min<sup>-1</sup> — any consistent unit is fine; the solidus form is not.

**Significant figures** (June 2026 01A (a)(ii)): "answer given to two significant figures" — 0.14–0.16 dm<sup>3</sup> per minute.

### Risks (Oct 2023 Q2(a); June 2025 Q4(a))

| Risk                                              | Reduction                                                  |
|---------------------------------------------------|------------------------------------------------------------|
| infection from the mouthpiece                     | <b>sterilise / replace the mouthpiece</b> after each use   |
| CO <sub>2</sub> build-up (soda lime exhausted)    | <b>replace the CO<sub>2</sub> absorber</b>                 |
| breathing difficulties / named condition (asthma) | <b>medical questionnaire</b> ; do not take part if at risk |
| soda lime is an <b>irritant</b>                   | gloves / goggles                                           |

"Ignore clean unqualified." Ethics: **informed consent**; "students cannot be forced to participate".

### Variables

- Biotic (Oct 2021, Oct 2023, June 2026 01A): **age, sex** ("IGNORE gender, ACCEPT sex" in 2026; Oct 2023 allowed gender — write **sex**), **body mass / BMI / height, fitness level, health / named disease e.g. asthma**, smoking, **playing a wind instrument, intensity / duration of exercise**.
- Abiotic: **room temperature, humidity, altitude, O<sub>2</sub> / CO<sub>2</sub> concentration, air quality**.

## Theory

- **Why breathing increases in exercise** (Oct 2021): more  $O_2$  for **aerobic respiration**; muscles need more **ATP** for contraction; more  **$CO_2$**  to remove;  $O_2$  for **oxidation of lactate**.
  - **Control of ventilation** (Oct 2023 Q2(d), 3): **chemoreceptors** detect  $\uparrow CO_2$  /  $\downarrow pH$  in blood  $\rightarrow$  **impulses** to the **ventilation centre** (medulla)  $\rightarrow$  to **intercostal muscles and diaphragm**  $\rightarrow$  rate and depth increase.
- 

## 13. Habituation

**Tested:** June 2021 plan (*Aplysia*), Oct 2022 plan (polychaete), June 2024 Q2 (woodlice; zebrafish sound), Jan 2025 Q3 (*Aplysia* data), Oct 2025 Q2 (Siamese fighting fish + mirror).

### Mark points (June 2024 Q2(a) woodlice; Oct 2025 Q2(b) fighting fish)

1. **Stimulus given in a standardised manner** — "eg glass rod / cotton bud / same force"; mirror of fixed size added to the tank.
2. **Repeated stimuli on the same animal** until **no response** is seen (or at least five times) — "Do not accept to calculate mean / SD" for this point.
3. **Record the number of touches** before no response / **time to uncurl** / **duration of the display** ("Ignore time for fins to extend").
4. **Stated interval between stimuli / let the animal rest** — **remove / cover the mirror** between trials.
5. **Repeat with different animals** of the same species.
6. **Control two variables** — temperature (**AC room** for woodlice; **thermostatic heater in the tank** for fish), light (bulb of known wattage / fixed distance), age / size of animal, **tank size / volume of water**, humidity.

### Why use a mirror and not another fish (Oct 2025 Q2(a))

"the animals may be harmed / fight" or "the other fish may become habituated / change its behaviour".

### Why use each animal only once (June 2022 Q1(d))

"so fish cannot learn / become habituated".

### Theory: why habituation is an advantage (Oct 2025 Q2(c); Jan 2025 Q3(e))

- **less energy used** on a response to a harmless stimulus;
- **other behaviours are possible** — **feeding**, reproduction, avoiding predators;
- **reduces risk of damage** to the fish / its fins;
- *Aplysia*: gills out longer  $\rightarrow$  **more oxygen uptake**.

## The data

- June 2024 zebrafish: percentage decrease = **71** ("Not 71.1 / 71.0"). Graph: a **line graph** of mean distance against stimulus number ("not MP3 if a bar graph").
- Jan 2025 *Aplysia*:  $r_s = 0.1$ , critical value **1.000** ( $n = 5$ ) → accept  $H_0$ .

## 14. Microbiology — aseptic technique, zones of inhibition, dilution plating, turbidity

**Tested:** Jan 2021 Q3 (*E. coli* ampicillin), Oct 2021 plan (pineapple), Jan 2023 Q2 (k), June 2023 plan (*Piper betle*), Jan 2024 Q1 (*Tussilago*), June 2025 Q3(d) (dilution plating), Jan 2026 Q2 (yeast turbidity), June 2026 01 plan (antibiotics).

### 14.1 Antimicrobial activity — the disc / well method (Jan 2024 Q1(a), 6)

1. Use **two or more plant parts** (root, stem, leaf, flower) / concentrations.
2. **Standardised extraction** — same mass, same volume of solvent, same grinding time.
3. Prepare **agar plate / broth with bacteria** (a lawn).
4. **Aseptic technique described** — flame the loop / forceps; sterilise dish; **autoclave**; disinfect bench; work **near a Bunsen flame**; lift the lid only slightly.
5. **Discs soaked in extract** (or wells) — plus a **control disc soaked in solvent only**.
6. **Incubate for a stated time and temperature** — **24–72 h, 20–30 °C**.
7. **Measure the diameter / area of the zone of inhibition** (clear zone).

### 14.2 The aseptic rules the schemes like

| Step                                                               | Reason                                                        |
|--------------------------------------------------------------------|---------------------------------------------------------------|
| <b>incubate at 25 °C (below 37 °C)</b>                             | human <b>pathogens</b> less likely to grow                    |
| tape the lid with <b>two short pieces</b> , not sealed all round   | <b>aerobic</b> conditions, so anaerobic pathogens do not grow |
| <b>flame</b> loop / <b>ethanol</b> on spreaders                    | kill microorganisms on equipment                              |
| work near a <b>Bunsen flame</b>                                    | rising air carries contaminants away                          |
| <b>autoclave</b> agar and dishes; dispose of plates by autoclaving | kill all microbes                                             |

The risk mark (Jan 2023 Q2(a)): "*harmful bacteria / infection / pathogens / biohazard — not contamination unqualified*"; control: good aseptic technique, gloves, **incubate below body temperature**. "*Emphasis is on risk to humans.*"

### 14.3 Dilution plating (*June 2025 Q3(d)*, 6 — each method point with its reason)

| Method                                                       | Reason                                                    |
|--------------------------------------------------------------|-----------------------------------------------------------|
| serially dilute the culture                                  | too many bacteria to count individual colonies            |
| spread a known volume on agar                                | to separate individual bacteria (each gives one colony)   |
| incubate for a fixed time / temperature (20–35 °C, 1–4 days) | so colonies grow big enough to see / count                |
| use aseptic technique                                        | to avoid contamination / competition / errors in counting |

Number in original = colonies × dilution factor ÷ volume plated. Count only plates with **30–300** colonies (separate colonies).

### 14.4 Growth-rate constant k (*Jan 2023 Q2(b)*)

$$k = (\log_{10} N_t - \log_{10} N_0) \div (\log_{10} 2 \times t) = (\log 14\,000 - \log 900) \div (0.301 \times 2) = (4.15 - 2.95) \div 0.602 = \mathbf{1.99} \text{ (3 s.f.)}$$

k is the number of **generations per hour**. Show both log values and the denominator — each is a mark.

### 14.5 Turbidity (*Jan 2026 Q2*)

- Measure turbidity with a **colorimeter** (absorbance / optical density / light transmission).
- **Same strain / age of yeast; standardised start** — same mass of yeast, concentration of sugar, volume of broth, temperature; **buffer**; **measure at each time interval**; **agitate / shake** the culture before sampling.
- Rate:  $(8.0 - 1.8) \div 4 = 1.55 \rightarrow \mathbf{1.6 \text{ a.u. h}^{-1}}$  (2 s.f.).
- Variables: **temperature, substrate concentration, oxygen concentration** ("Ignore dead cells").

### 14.6 Theory marks

- **Bacteriostatic** — prevents increase in numbers; **bactericidal** — kills bacteria (*Jan 2024*).
- **Prescribing code** (*Jan 2024*): only for **bacterial** infections / not for viral / not for minor infections / **finish the course** → **reduces antibiotic resistance** ("allow creates selection pressure").

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## 15. Vitamin C with DCPIP

**Tested:** June 2022 Q2 (juice), Oct 2023 Q1 (camu camu storage).

DCPIP is **blue**; vitamin C (a reducing agent) **decolourises** it. Titrate juice into a fixed volume of DCPIP.

### Mark points (*June 2022 Q2(a)* and *Oct 2023 Q1(b)*)

1. **DV = volume of juice needed to decolourise a fixed volume of DCPIP** (*"Ignore amount unqualified"*).
2. **Making a juice extract** — crush / blend / pestle and mortar / filter — by a **standardised method**.
3. **Control** — same **volume / concentration of DCPIP**, same **mass of fruit / volume of juice**.
4. **Colour change: blue → colourless** (or the juice colour / pink).
5. **Record the volume (or number of drops) to reach the end point** — "titrate".
6. **Calculate vitamin C concentration** — **calibration curve** of known vitamin C concentrations /  $C_1V_1 = C_2V_2$ .
7. **Repeats and means / SD**.
8. (Storage version) store fruits at the **same temperature** for each time; another storage variable — **humidity**, mass, source, age.

*"Ignore pH / temperature / age of fruit"* for the control mark in June 2022.

### Data points

- Percentage change (*June 2022*):  $(223.3 - 92.7) \div 92.7 \times 100 = \mathbf{141\%}$  — *"Answer must be to 3 sig figs."*
- Percentage (*Oct 2023*): **23**.

### Theory

- **Blood clotting** (*June 2022, 4, in order*): **thromboplastin** released → **prothrombin** → **thrombin** → **fibrinogen** → **fibrin** → **fibrin traps red blood cells / platelets**. *"MAX 3 if one statement is in the wrong order."*
- **Collagen** (*Oct 2023, 3*): **polypeptide**; **triple helix**; held by **hydrogen bonds**; **fibrous**; **glycine every third amino acid**. *"Not beta pleated sheet."*

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## 16. Water potential of plant tissue

**Tested:** Oct 2020 Q1 (potato).

### Mark points (*Oct 2020 Q1(b)*)

1. **Same initial mass / volume / length** of cylinders (cork borer + scalpel + ruler) — *"IGNORE size"*.
2. **Five concentrations** of sucrose (or salt) solution.
3. **Temperature controlled / stated** — *"ALLOW air-conditioned room"*.

4. **Measure the change in mass / length after the same / stated time (30 min – 3 days);** blot dry before weighing.
5. **Repeats and mean.**
6. **Plot percentage change in mass against concentration; read where the line crosses the x-axis** (zero change) → the solution has the **same water potential** as the tissue.

Use **percentage** change in mass, because the starting masses differ slightly.

Variables (Oct 2020 Q1(c)): abiotic — pH, temperature, starting mass; biotic — **same age / same part of tuber**. Effect: *"IGNORE accuracy / precision / reliability"; "ALLOW description of how membranes affected"*.

Why species differ (Oct 2020 Q1(d)): different **solutes**, different **solute concentrations** → different **water potential**; **cell walls** differ (**turgor / pressure potential**).

Starch (Oct 2020 Q1(a)): **polymer of glucose**; large → **insoluble / no osmotic effect**; **branched** → **hydrolysed quickly**. *"IGNORE easily."*

## 17. Tensile strength of plant fibres

**Tested:** Jan 2022 Q1 (kenaf), Oct 2024 Q2 (kenaf retting).

**Mark points (Jan 2022 Q1(a), 6)**

1. **DV = mass needed to break the fibre.**
2. **Same length and diameter / cross-sectional area** of fibres.
3. **Method of supporting the fibre** — clamped between two stands / hung from one.
4. **Add known masses** until the fibre breaks (or use a **forcemeter**).
5. **Control one variable** — **age of fibre / plant, temperature** (*"Accept AC room for temperature (not waterbath)"*), **humidity** (AC room / humidifier / dehumidifier).
6. **Tensile strength = force ÷ cross-sectional area** (mass × 9.8 → force).
7. **Repeats and mean.**

**Risk (Oct 2024 Q2(a))**

**Falling masses** → cushion / sand tray underneath, stand back; **fibre snapping into the eye** → **goggles / safety screen**; plant allergies → gloves; **hydrogen peroxide is an irritant** → gloves.

### Theory

- **Cellulose** (Jan 2022, June 2023, 3): **β-glucose**; **1–4 glycosidic bonds**; **every other glucose inverted**; **straight, unbranched chain**. *"Ignore cellulose molecules joining together."*
- **Sclerenchyma fibres** (Oct 2024, 2): long cells with **lignin / secondary thickening**; **end walls present**; **dead / hollow**; **pits**.

- **Biofuels** (*Jan 2022*): release only the CO<sub>2</sub> that was **absorbed in photosynthesis** — **carbon neutral**.
  - Oct 2024 percentage increase: mean 834 → **10.4%** (3 s.f.).
- 

## 18. Mitotic index — root tip squash

**Tested:** June 2023 Q1 (caffeine), Oct 2024 Q3 (oxygen, data).

### Mark points (*June 2023 Q1(b)(i)*, 6)

1. **Treatment:** onion roots / bulbs grown in **caffeine solution**.
2. **With and without caffeine** (0% as control) / stated concentrations.
3. **Use the root tip** — "*e.g. terminal 5 mm*".
4. **Named stain** — **toluidine blue / orcein / methylene blue**.
5. **Slide preparation** — **treat with acid** (HCl, to separate cells), **macerate / tease apart / squash** under the coverslip.
6. **Count cells** under the microscope — cells in mitosis and total cells in the field of view.
7. **Mitotic index = number of cells in mitosis ÷ total number of cells** (× 100 for a percentage).

Stage identification (*June 2023 Q1(a)*): chromatids pulled to opposite poles = **anaphase** ("*Ignore ref to early / late anaphase*").

Data (*Oct 2024 Q3*): means **5.7 and 4.0** ("*Do not allow 4*" — keep the decimal place),  $t = 11.3 > 2.12$  → reject H<sub>0</sub>; "*pea seeds grown under 20% oxygen have a larger mitotic index than those grown under 5% oxygen*".

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## 19. Seeds, gibberellin and starch agar; pollen tubes

### 19.1 Gibberellin and amylase — starch agar (*Oct 2024 Q1*, 6)

1. **Aseptic technique** — wash grains with **sterile water / sodium hypochlorite**; sterile plate.
2. **Cut the seed in half** to separate the **embryo** from the **endosperm**.
3. **Soak both halves in gibberellin**.
4. **Control: halves soaked in water**, not gibberellin.
5. Place halves **cut side down on a starch-agar plate**.
6. **Incubate for 24–48 h at a stated / fixed temperature** ("*Do not allow waterbath*").
7. **Add iodine** after incubation — the **clear zone** (no blue-black) shows where starch was digested.
8. **Compare** the diameter / area of the clear zones. "*Ignore zone of inhibition*" — it is a clear zone of **digestion**, not inhibition.

Theory (Oct 2024 Q1(c)): **gibberellin is (acts as) a transcription factor → increases transcription of amylase mRNA → translation of amylase.** (*"Ignore genes / cAMP"*.) Amylase converts starch to maltose.

## 19.2 Pollen tube growth (Jan 2021 Q1, Jan 2024 Q2)

Mark points (Jan 2021 Q1(a), 6):

1. **Compare pollen with and without the stigma extract** (sucrose-only as the control; *"different concentrations ... as long as one is zero"*).
2. **Measure the pollen tube with a microscope — eyepiece graticule calibrated with a stage micrometer** (*"Need to mention microscope"*).
3. **Control one variable** — temperature, pH, volume of solution, light intensity, age of plant / pollen.
4. **Control a second variable.**
5. **Stated time intervals** — e.g. every 10–30 min, or a single interval between 30 min and 12 h.
6. **Rate = change in length ÷ time.**
7. **Repeats and mean / SD.**

Data (Jan 2024 Q2(a)): **20.0  $\mu\text{m h}^{-1}$**  — the **unit is a mark** ( $\mu\text{m hour}^{-1}$ ,  $\mu\text{m per hour}$ ).

Theory — how the tube tip grows (Jan 2021 Q1(b), 4): starch **hydrolysed** to glucose; **condensation** reactions form **pectin**; pectin in **vesicles** (Golgi); vesicles **fuse with the membrane at the tip** (**exocytosis**); **mitochondria** supply **ATP**.

Temperature theory (Jan 2024 Q2(b)): higher temperature → more **enzyme activity / ES complexes** → more **respiration** → faster growth; very high → **denature**.

Variables (Jan 2024 Q2(c)): **mineral / sucrose / boron concentration** of the growth medium, **pH**, **light intensity**, **humidity**. (*"Ignore nutrients / oxygen / carbon dioxide."*)

## 19.3 Seedling / plant growth — *Lemna* (Jan 2023 Q1)

1. **DV** — number of plants / leaves (fronds) or **mass** (*"Ignore rate"*).
2. **Five nitrate concentrations.**
3. **Suitable apparatus** — **Petri dish / beaker** so plants can be counted — *"not test tube"*.
4. **Stated time** — **minimum 1 day**.
5. **Control one variable** — temperature (**thermostatic water bath / AC / incubator**), pH (**buffer**), light (**bulb at set distance**).
6. **Repeats and means.**
7. **Rate** — number or mass ÷ time.

Theory (Jan 2023 Q1(b); Jan 2025 Q2(c)): nitrate (nitrogen) is used to make **amino acids / proteins** → **enzymes (RUBISCO)**; **nucleotides / DNA / RNA** → **cell division**; **chlorophyll** → photosynthesis → **GALP / sugars**.

## 20. Brine shrimp hatching — ions and enzymes

**Tested:** June 2021 Q1 (calcium), Jan 2025 Q1 (copper).

**Mark points** (*June 2021 Q1(a)*, *Jan 2025 Q1(a)*)

1. **DV = number of eggs hatched per unit time / in a stated time** (or time for a fixed number to hatch).
2. **Five concentrations** of the ion — **include zero**.
3. **Count shrimps hatched / swimming at same / stated intervals** — single count: 1 h to 3 days.
4. **Control of one variable** — **temperature** (thermostatic water bath, up to 45 °C), **pH** (buffer), **light intensity**, **salinity** (stated g dm<sup>-3</sup>), number / age of eggs, species.
5. **A second variable** (2025).
6. **Repeats and means / SD**.
7. **Further detail** — how eggs are delivered (same mass on a spatula), making the dilutions, shining a light to attract shrimps for counting.

Ethics (*Jan 2025 Q1(b)*): **no ethical concern** because shrimps are **not vertebrates / cannot feel pain / are not harmed / are plentiful**. "*Ignore simple nervous system.*" (Contrast *Jan 2022 Q2(a)*, where "snails have a simple nervous system, so thought not to feel (much) pain" **was** accepted — always link it to **pain**.)

Theory:

- **Calcium ions as cofactor** (*June 2021*): allow **substrate to bind to the active site**; **lower activation energy**; faster **respiration**; faster **anabolic** reactions → earlier hatching.
- **Copper as inhibitor** (*Jan 2025*): Cu<sup>2+</sup> interferes with **ionic bonds / R groups** → alters **tertiary structure** → **active site shape changed** → **fewer ES complexes**.

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## 21. Ecological sampling and growth measurements

### 21.1 Sampling methods — which one to use

| Situation                                      | Method                      | Key mark words                                                                                                                                              |
|------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| comparing <b>two areas</b>                     | <b>random sampling</b>      | <b>grid</b> , co-ordinates from a <b>random number generator</b> ( <i>June 2025</i> ; " <i>Ignore select 5 sample sites at random</i> " — you must say how) |
| a <b>gradient</b> (dune, shore, woodland edge) | <b>belt / line transect</b> | <b>quadrats at regular intervals</b> along a transect <b>at 90° to the shoreline</b> ( <i>Jan 2026</i> )                                                    |

| Situation            | Method                                                         | Key mark words                                              |
|----------------------|----------------------------------------------------------------|-------------------------------------------------------------|
| mobile invertebrates | <b>pitfall traps</b> , sweep net, kick sampling                | same number of traps, same time left ( <i>Jan 2026 Q3</i> ) |
| plants               | <b>quadrat</b> : count, <b>percentage cover</b> , <b>ACFOR</b> | "Accept up to 5 m <sup>2</sup> " for quadrat size           |

## 21.2 Transect mark points (*Jan 2026 Q1(b)*, 5)

1. **Transect line**.
2. **Quadrat**.
3. **Defined quadrat size**, or transect at **90° to the shoreline**.
4. **Standardised placing** — regular intervals / stated distance.
5. **Count plants / percentage cover / ACFOR**.
6. **Measure one abiotic variable** — pH or salinity of sand, humidity, temperature, **wind speed**.
7. **Repeat with at least one more transect**.

## 21.3 Field-study risks (*Oct 2021, Jan 2026, Oct 2019*)

slips and trips → suitable footwear; **sunburn / heatstroke** → sunscreen, hat, water; **bites / stings** → insect repellent, long sleeves; **allergy to plants** → gloves; **sand / salt in eyes** → goggles; **strangers / weather** → **do not work alone**, check the forecast. ("*gloves qualified*" — say what they protect against.)

## 21.4 Soil abiotic variables (*June 2023, Jan 2025, June 2025*)

**pH, water content / soil moisture, temperature, mineral content / salinity, organic matter / humus, compaction, depth, O<sub>2</sub> / CO<sub>2</sub> content.**

**Jan 2025 ruling:** "*Reject light / humidity*" as soil variables — they are air variables. "*Ignore nutrients*" (*June 2025*) — say **minerals** or **nitrate**.

Control in a field study (*June 2025 Q2(c)(ii)*): you cannot control the field, so **select sample sites with similar values** of the variable (measure pH first and only use sites within a range). "*not pH and buffer*" in the field.

## 21.5 Growth measurements — maggots, earthworms, lizards

- **Maggot growth** (*Oct 2025 Q1(c)*, 3): **maggots hatched at the same time**; **measure the length of a sample every 24 h** ("*any time apart from 24 hours*" ignored); **several maggots** → **mean**; both types of meat. Maximum length reached at **168 h**; growth rate **0.33 mm h<sup>-1</sup>** — **unit required**.
- **Earthworms** (*June 2025 Q2*): **magnification = image size ÷ actual size** → 0.37–0.5 (the image was smaller than life). **Standard deviation = 2.2**. Comparing two populations: **compare the means**, then use the **SDs** — **t-test or look for overlap** ("*Ignore comment on anomalies / validity / reliability*").

- **Animal variables** (Oct 2025 Q1(b)): temperature, **oxygen concentration** (air-line / gas supply in incubator), **humidity** (AC room), **mass of food** (balance), light intensity / wavelength (same bulb / filter).

### 21.6 June 2026 (01) Q1 — flour beetles (*model answer — no mark scheme*)

- **(a) Actual length** between A and B: measure the image in mm with a ruler, then **actual length = image length ÷ 15**. (The value depends on the printed photograph — do the step in the exam, and give the answer in **mm**.)
  - **(b) Optimum temperature for the number of adults (6)** — built from the template in §8: 1. DV = **number of adult beetles** emerging after a **stated time** (e.g. 30 days). 2. **Five temperatures**, e.g. 20, 25, 30, 35 and 40 °C, in **incubators**. 3. **Same number of eggs** (e.g. 20) in each container — counted under a hand lens. 4. **Same mass and type of flour / cereal** (balance). 5. Control **humidity** (humidifier in the incubator) and **light** (dark incubator). 6. **Repeats** at each temperature and **mean**; the optimum is the temperature with the highest mean number of adults; then **repeat with smaller intervals** around it.
  - **(c) Resistance to insecticide (3)**: a random **mutation** gives an allele for resistance → the insecticide is a **selection pressure** → resistant beetles **survive and reproduce** → pass on the allele → **allele frequency increases** over generations.
-

## PART C — VARIABLES, RISKS AND ETHICS

### 22. Variables — the 3–5 marks on every paper

#### 22.1 The three question shapes

| Question                                                              | Marks | What scores                                                                                                                                                                                       |
|-----------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "State one abiotic and one biotic variable" / "two abiotic variables" | 1–2   | the <b>right category</b> — a biotic answer on an abiotic line scores 0 and can <b>negate</b> the line ("If biotic and abiotic factor stated on the same line they negate each other" — Oct 2022) |
| "Describe how one of these variables could be controlled"             | 1     | <b>variable + method</b> — "variable with suitable control method described"                                                                                                                      |
| "...and the effect on the results if it were not controlled"          | 1     | "results are not valid / description of expected effect on the dependent variable" — <b>directional</b>                                                                                           |

**The list rule:** "Mark the first answer on each line"; "Only mark the first response on each line". Write **one** variable per line. A wrong first answer is not rescued by a right second one.

**An inappropriate variable can still earn the control mark** — "An inappropriate biotic / abiotic variable can still gain both marks" (Jan 2023); "Allow ecf for inappropriate variable but correct method of control" (Oct 2025).

#### 22.2 The control-method table (from the schemes' ACCEPT lists)

| Variable                                 | Control method the schemes accept                                       | Notes                                                                                                               |
|------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>temperature</b> (lab, in solution)    | <b>thermostatic / thermostatically controlled water bath, incubator</b> | "Ignore waterbath unqualified" (Jan 2026) — say <b>thermostatic</b>                                                 |
| <b>temperature</b> (room / whole animal) | <b>air-conditioned (AC) room, thermostatically controlled room</b>      | for fibres: "Accept AC room ... (not waterbath)"; for snails: "ignore waterbath unless water temperature specified" |
| <b>temperature</b> (fish tank)           | <b>thermostatic heater in the tank</b>                                  |                                                                                                                     |

| Variable                                          | Control method the schemes accept                                                                       | Notes                                                                         |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>temperature</b> (aquatic plant)                | thermostatic water bath; <b>heat shield</b> between lamp and plant                                      | <i>"Ignore AC room"</i>                                                       |
| <b>pH</b>                                         | <b>buffer</b> (stated pH)                                                                               | in the <b>field</b> : <i>"not pH and buffer"</i> — select sites of similar pH |
| <b>light intensity</b>                            | <b>same wattage bulb at a fixed / measured distance</b> ; light meter                                   |                                                                               |
| <b>wavelength</b>                                 | <b>same filter / same coloured bulb</b>                                                                 |                                                                               |
| <b>humidity</b>                                   | <b>AC room, humidifier / dehumidifier</b>                                                               |                                                                               |
| <b>oxygen concentration</b>                       | <b>air-line / gas supply</b> into the incubator                                                         | <i>Oct 2025</i>                                                               |
| <b>mass</b> of organism / food / tissue           | <b>balance</b> — same mass                                                                              |                                                                               |
| <b>length / size</b> of pondweed, fibre, cylinder | <b>ruler, cork borer</b>                                                                                | <i>"IGNORE size"</i> — say <b>length / mass / diameter</b>                    |
| <b>age</b> of organisms                           | <b>hatched at the same time</b> , seeds from the <b>same batch</b> , same length / mass as an age proxy |                                                                               |
| <b>species / variety</b>                          | <b>use a key</b> , same supplier, same parent plant                                                     |                                                                               |
| <b>sex</b> of animal                              | select animals of <b>known sex</b>                                                                      |                                                                               |
| <b>salinity</b>                                   | stated % or g dm <sup>-3</sup>                                                                          |                                                                               |
| <b>volume / concentration</b> of a solution       | <b>same volume</b> measured with a <b>pipette / syringe / measuring cylinder</b>                        |                                                                               |
| <b>sound</b> (habituation)                        | <b>fixed duration</b> (e.g. 1 s), <b>fixed distance</b>                                                 |                                                                               |
| <b>stimulus</b> (touch)                           | <b>same glass rod / cotton bud, same force, same person</b>                                             |                                                                               |

## 22.3 The directional effect — model sentences

The schemes: "Description of effect must be directional eg increase temperature increases enzyme activity" (June 2023); "Accept an increase or decrease but not just affected / changed" (June 2022); "IGNORE accuracy / precision / reliability / repeatability" (Oct 2020).

| Variable        | Directional effect sentence                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------------------|
| temperature     | "A <b>higher</b> temperature would <b>increase</b> enzyme activity, so <b>more</b> oxygen would be produced per minute." |
| light intensity | "A <b>brighter</b> light would <b>increase</b> the rate of photosynthesis, so <b>more</b> bubbles."                      |
| mass of seeds   | "A <b>greater</b> mass of seeds would use <b>more</b> oxygen, so the fluid would move <b>further</b> ."                  |
| pH              | "A pH <b>further from the optimum</b> would <b>reduce</b> enzyme activity, so hatching would be <b>slower</b> ."         |
| age of animal   | " <b>Older</b> woodlice may <b>habituate faster</b> , so <b>fewer</b> touches before no response."                       |
| humidity        | " <b>Lower</b> humidity would <b>increase</b> water loss, so the snails would be <b>less active</b> ."                   |
| fitness         | "A <b>fitter</b> volunteer would have a <b>larger</b> tidal volume, so <b>higher</b> minute ventilation."                |

The safe alternative is "*the results would not be valid*" — it is on the mark scheme list every time. **Do not** write "not reliable" / "not accurate" — they are ignored.

## 22.4 Abiotic vs biotic — the banks by context

| Context                   | Abiotic                                                                                                                   | Biotic                                                                                                          |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| seeds / respirometer      | temperature, pH, water, CO <sub>2</sub>                                                                                   | variety / age / mass of seed, stage of germination, presence of microorganisms                                  |
| maggots / insects         | temperature, <b>oxygen</b> , <b>humidity</b> , light intensity / wavelength                                               | <b>mass</b> , <b>species</b> , <b>age</b>                                                                       |
| fish                      | <b>water temperature</b> , <b>light intensity / wavelength</b> , <b>pH</b> , <b>oxygen concentration</b> , tank size      | <b>age</b> , <b>sex</b> , <b>mass</b>                                                                           |
| toad skin / animal tissue | temperature, pH                                                                                                           | <b>age / sex</b> of toad, <b>part of body /</b> thickness of skin, disease                                      |
| human breathing           | <b>room temperature</b> , <b>humidity</b> , <b>altitude</b> , O <sub>2</sub> / CO <sub>2</sub> concentration, air quality | <b>age</b> , <b>sex</b> , <b>BMI / mass / height</b> , <b>fitness</b> , <b>health / named disease</b> , smoking |

| Context          | Abiotic                                                                                            | Biotic                                                               |
|------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| soil / field     | pH, water content, temperature, mineral content / salinity, organic matter, compaction             | species, age of plant, <b>competition</b> , <b>grazing</b> , disease |
| bacteria / yeast | temperature, pH, oxygen, growth medium / substrate concentration                                   | <b>strain / species</b> , age of culture, starting density           |
| pollen tubes     | mineral / sucrose / boron concentration, pH, light, humidity                                       | age of pollen, species                                               |
| plant fibres     | pH / water content / organic matter of soil, humidity, H <sub>2</sub> O <sub>2</sub> concentration | <b>age, species / variety, diameter / length</b> of fibre            |

**Things the schemes ignore:** "*nutrients*" (say **minerals** / nitrate / glucose), "*size*", "*type*", "*genetic variation*", "*disease*" (for maggots), "*stress*", "*dead cells*".

## 22.5 Variables that cannot be controlled (field work)

June 2026 01 Q2(c)(ii) — *model answer, no mark scheme*: variables that cannot be controlled are **measured / monitored** so that **their effect can be taken into account when drawing conclusions** — you can check whether they were similar at each site, and whether they, rather than the IV, could explain a difference.

## 23. Risks — the 2-mark "risk + reduction" pair

**Mark scheme shape:** "*suitable risk identified (1) + suitable control / reduction identified (1)*". "*Mark first risk identified.*" The reduction must **match** the risk.

| Practical       | Risk                                                                                 | Reduction                                                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| microbiology    | infection by pathogens / harmful bacteria (" <i>not contamination unqualified</i> ") | <b>aseptic technique</b> ; lid taped with two strips, <b>not opened</b> ; <b>incubate below 30 °C</b> ; gloves; <b>autoclave</b> after use |
| spirometer      | infection from mouthpiece                                                            | <b>sterilise / replace mouthpiece</b>                                                                                                      |
|                 | CO <sub>2</sub> build-up                                                             | <b>replace soda lime</b>                                                                                                                   |
|                 | asthma / heart condition                                                             | <b>medical questionnaire</b> ; exclude at-risk volunteers                                                                                  |
| soda lime / KOH | <b>corrosive / irritant</b>                                                          | <b>gloves, goggles</b>                                                                                                                     |

| Practical                                              | Risk                                                                                                                | Reduction                                                                         |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| plant extracts /<br>arsenic / pyrethrum /<br>chemicals | toxic, irritant, allergy                                                                                            | gloves, mask ("ignore infection /<br>labcoat / PPE" — Oct 2022: name the<br>item) |
| fibre / tensile test                                   | falling masses; snapping fibre hits<br>eye                                                                          | cushion / sand tray underneath, stand<br>back; goggles / safety screen            |
| H <sub>2</sub> O <sub>2</sub> (retting)                | irritant                                                                                                            | gloves, goggles                                                                   |
| field work                                             | slips, trips, bites, stings, sunburn,<br>heatstroke, allergy, spines,<br>strangers, weather, sand / salt in<br>eyes | footwear, repellent, sunscreen, water,<br>do not work alone, goggles              |
| scalpel / cork borer                                   | cuts                                                                                                                | cut away from the body on a tile                                                  |
| hot water bath                                         | scalds                                                                                                              | handle with tongs; check temperature                                              |

**Common loss:** writing "wear a lab coat" or "be careful". The reduction must be **specific** to the named risk.

## 24. Ethics — animals and humans

### 24.1 Animals

| Scheme wording accepted                                                                                                                                                                                | Paper     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| zebrafish <b>not caused pain / returned to the source</b> to prevent ecological change / <b>humane conditions</b> — <i>"Ignore ... simple nervous system or does not feel (much) pain unqualified"</i> | June 2022 |
| snails have a <b>simple nervous system, so thought not to feel (much) pain</b> / invertebrates / <b>no licence needed</b>                                                                              | Jan 2022  |
| brine shrimp — <b>not vertebrates / cannot feel pain / not harmed / plentiful</b> — <i>"Ignore simple nervous system"</i>                                                                              | Jan 2025  |
| each fish used <b>once</b> — so it cannot <b>learn / become habituated</b>                                                                                                                             | June 2022 |
| mirror instead of a second fish — <b>animals may fight / be harmed</b>                                                                                                                                 | Oct 2025  |

| Scheme wording accepted                                                                                               | Paper     |
|-----------------------------------------------------------------------------------------------------------------------|-----------|
| habituation — <b>sound not too loud, rest time between tests, don't test the same fish many times</b> , suitable tank | June 2024 |

**The rule:** link the reason to **pain** or **harm**, or to **returning the animal / the population**. "They are small" or "it's legal" scores nothing.

## 24.2 Humans (spirometer, fitness)

- **Informed consent** — volunteers told the risks and can **withdraw**; *"students cannot be forced to participate"*.
- **Health screening** — medical questionnaire; do not exercise if at risk.
- **Confidentiality** of personal data.
- Hygiene of shared equipment.

## 24.3 Plants and the environment

- Use a species that is **not endangered**; take only **small samples**; **replace** turf / return organisms; do not **trample** (*Jan 2026 Q4(a): "the plant is not an endangered species"*).

## PART D — THE DATA-ANALYSIS QUESTION

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### 25. The shape of the data question

Q3 (or Q2 in three-question papers) gives raw data from an investigation and walks you through the same sequence almost every time:

| Step                                                  | Marks           | Section |
|-------------------------------------------------------|-----------------|---------|
| (a) <b>null hypothesis</b>                            | 1 (Oct 2019: 2) | §26     |
| (b) <b>table</b> of the raw data + means (or medians) | 2–3             | §27     |
| (c) <b>graph</b> of the means with range bars         | 2–3             | §28     |
| (d)(i) <b>statistical test</b> — calculate            | 2–3             | §29–§33 |
| (d)(ii) <b>conclusion</b> from the test (and graph)   | 2–4             | §34     |
| (e) <b>extend / improve</b> the investigation         | 2–3             | §37     |

That is **12–15 marks** that are almost entirely technique. They are the most reliable marks on the paper.

---

### 26. The null hypothesis

#### The template

"There is no (significant) difference between the (mean) {DV} of {group A} and {group B}."  
"There is no (significant) correlation between {variable 1} and {variable 2}."

#### The rulings

| Ruling                                                                                                                    | Paper                                                           |
|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| a <b>difference</b> test (t-test, Mann–Whitney) needs "no difference" — <b>"DO NOT ACCEPT correlation / relationship"</b> | Oct 2019; Jan 2021: "Ref to correlation & difference = no mark" |
| a <b>correlation</b> test (Spearman) needs "no correlation"                                                               | Oct 2021, Oct 2022, Jan 2025, June 2025, June 2026 01A          |
| <b>name the DV and both groups</b> — "Three parts to this answer"                                                         | Jan 2021                                                        |

| Ruling                                                                                                                               | Paper               |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| the groups must be <b>qualified</b> — "A and B must be qualified"<br>("before and after the hurricane", not "sample A and sample B") | Jan 2024            |
| "Ignore fields A, B and C unqualified"                                                                                               | Oct 2025            |
| "Ignore effect" — say "no difference in biomass", not "cows have no effect"                                                          | Oct 2025            |
| $\chi^2$ : "there is <b>no difference between the observed and expected numbers</b> " / no preference                                | Jan 2025, June 2024 |
| Spearman: "Accept the correlation coefficient ... will be 0"                                                                         | June 2025           |

### Which one?

| Design                                                                                  | Hypothesis                                         | Test                  |
|-----------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------|
| two groups, continuous data, normal (lengths, masses, times, counts of many)            | <b>no difference</b>                               | <b>t-test</b>         |
| two groups, data <b>not normally distributed</b> / small ordinal data                   | <b>no difference</b> (in medians)                  | <b>Mann–Whitney U</b> |
| one variable changed in <b>several steps</b> against another continuous variable        | <b>no correlation</b>                              | <b>Spearman</b>       |
| <b>categories / counts</b> in groups (choice chambers, fruiting / not fruiting, ratios) | <b>no difference between observed and expected</b> | $\chi^2$              |

Why  $\chi^2$  (Jan 2025 Q2(b)(iii)): "the data are categorical / nominal" — "Accept not a normal distribution / not continuous data / raw counts / frequency data".

## 27. The table — 3 marks

### Mark points

1. "suitable table format with data" — raw data in **columns**, one column per group.
2. "correct column headings with units" — **quantity / unit** in the heading **only**.
3. "means correctly calculated" — to the **same number of decimal places as the raw data**.

## Rulings

| Ruling                                                                                                                                  | Paper                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| "Do not accept units in the body of the table"                                                                                          | June 2021; Oct 2024:<br>"not if units in table" |
| "A third column negates MP1" / "MP1 do not accept an extra column"                                                                      | Jan 2024; Jan 2023                              |
| "Reject incorrect columns" (a trap-number column was <i>ignored</i> in Jan 2026 — do not rely on it)                                    | Jan 2026                                        |
| means <b>to the same number of places as the data</b> — "Not 17 / 18 as mean values"; "Means can be to two places or one but not mixed" | June 2022; Jan 2021                             |
| "Do not allow 4" (for 4.0)                                                                                                              | Oct 2024                                        |
| Mann–Whitney: data <b>ranked from lowest to highest</b> , <b>medians</b> identified                                                     | Oct 2019; June 2021                             |

## The model table (Jan 2024 Q3(b))

| Surface area of toepads / mm <sup>2</sup> — before hurricane (A) | Surface area of toepads / mm <sup>2</sup> — after hurricane (B) |
|------------------------------------------------------------------|-----------------------------------------------------------------|
| 0.9                                                              | 0.8                                                             |
| 1.3                                                              | 1.3                                                             |
| 1.1                                                              | 1.5                                                             |
| 1.2                                                              | 1.8                                                             |
| 1.3                                                              | 1.3                                                             |
| 1.5                                                              | 1.5                                                             |
| 1.6                                                              | 1.6                                                             |
| 0.8                                                              | 1.9                                                             |
| 0.7                                                              | 1.8                                                             |
| <b>Mean 1.2</b>                                                  | <b>Mean 1.5</b>                                                 |

(The scheme accepted 1.16 / 1.50 or 1.2 / 1.5.)

**Draw the table with a ruler**, headings in the top row, the unit after a solidus (**1 mm<sup>2</sup>**), the mean in a clearly labelled final row. No "trial number" column.

## 28. The graph — 3 marks

### Two groups → bar graph

1. "bar graph with linear scale and axes labelled with units" — y: mean {DV} / unit; x: the two groups by name.
2. "means plotted correctly" (ECF from your table).
3. "range bars plotted correctly" — from the lowest to the highest value in each group.

| Ruling                                                                                                               | Paper                                                                                            |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>"Must start at zero ie not broken axis"</b>                                                                       | Jan 2022; Jan 2023 ("starting at zero without a break"); Jan 2024, Oct 2024, Jan 2026, June 2021 |
| the y-scale must <b>extend beyond the highest range bar</b> — "No mark can be awarded if linear scale stops at 22"   | Oct 2019                                                                                         |
| <b>separate bars of the same width</b>                                                                               | Jan 2025                                                                                         |
| plot <b>medians</b> , not means, after a Mann–Whitney table ("means plotted instead of median values" lost the mark) | Oct 2019; June 2021                                                                              |
| "Allow tolerance of half a small square" for range bars                                                              | Oct 2024                                                                                         |

(Jan 2021 accepted a broken scale, but later schemes did not. **Always start the bar-graph axis at zero.**)

### Continuous IV → scatter / line graph

1. **Axes labelled with units**, IV on x, **linear scale**.
2. **Points plotted correctly** in **scatter** format.
3. **Line of best fit** (June 2025, June 2026 01A) or **error bars** (Oct 2022) or **points joined + key** (June 2024).

| Ruling                                                                                                     | Paper                                            |
|------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| scatter axis <b>may start above zero</b> — "Accept broken scale with symbol or scale not starting at zero" | Oct 2022; June 2026 01A: "can start at non-zero" |
| <b>"DO NOT ACCEPT extrapolation"</b> / "Do not accept if line extrapolated to origin"                      | June 2026 01A; Jan 2025                          |
| a line graph for repeated stimuli — "not MP3 if a bar graph"                                               | June 2024                                        |

| Ruling                                                      | Paper              |
|-------------------------------------------------------------|--------------------|
| "Ignore line if drawn" on a scatter graph used for Spearman | Oct 2021, Oct 2022 |

## Graph must match the design

*bar graph + t-test for **two groups**; scatter / line graph + Spearman for a **continuous IV** (planning mark scheme, Oct 2023, Oct 2024, June 2025, Jan 2026).*

When there are **three** groups with a numeric IV (Oct 2025: 0, 2, 4 cows) the scheme accepted **either** a bar graph **or** a scatter graph.

## 29. Student's t-test

### The formula (as printed)

$$t = |\bar{x}_1 - \bar{x}_2| \div \sqrt{(s_1^2/n_1 + s_2^2/n_2)} \text{ degrees of freedom} = (n_1 - 1) + (n_2 - 1) = n_1 + n_2 - 2$$

The paper usually **gives**  $s_1^2$  and  $s_2^2$  (written  $(s_A)^2$  and  $(s_B)^2$ ). You calculate the **means** (from your table), the **difference**, the **denominator**, and **t**.

### Worked example — June 2026 (01) Q3, trained vs untrained volunteers (*model answer — no mark scheme*)

Resting O<sub>2</sub> consumption / dm<sup>3</sup> per minute: A (trained) 0.24, 0.25, 0.21, 0.18, 0.18; B (untrained) 0.22, 0.20, 0.19, 0.22, 0.18. Given  $(s_A)^2 = 0.00107$ ,  $(s_B)^2 = 0.00032$ .

| Step        | Working                                                                                                                                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| means       | $\bar{x}_A = 1.06 \div 5 = \mathbf{0.212}$ ; $\bar{x}_B = 1.01 \div 5 = \mathbf{0.202}$                                                                                                                                 |
| numerator   | $0.212 - 0.202 = \mathbf{0.010}$                                                                                                                                                                                        |
| denominator | $\sqrt{(0.00107/5 + 0.00032/5)} = \sqrt{(0.000214 + 0.000064)} = \sqrt{0.000278} = \mathbf{0.0167}$                                                                                                                     |
| t           | $0.010 \div 0.0167 = \mathbf{0.60}$                                                                                                                                                                                     |
| df          | $(5 - 1) + (5 - 1) = \mathbf{8} \rightarrow$ critical value at $p = 0.05 = \mathbf{2.31}$                                                                                                                               |
| conclusion  | $0.60 < \mathbf{2.31}$ , so <b>accept the null hypothesis</b> : there is <b>no significant difference</b> in resting O <sub>2</sub> consumption between trained and untrained volunteers; the <b>range bars overlap</b> |

(You can check the given variances: for A the squared deviations from 0.212 add to 0.00428;  $\div (5 - 1) = 0.00107$ . ✓)

### The mark scheme structure (e.g. Jan 2022 Q3(d)(i), 3 marks)

1. **correct numerator** —  $36.0 - 31.6 = 4.4$ ;
2. **correct substitution** of  $s^2/n$  values;
3. **correct t** — 1.03.

"Correct answer with no working gains full marks", but **ECF** (error carried forward) is only possible if the working is shown. "Ignore minus sign." "If  $(SA)^2$  and  $(SB)^2$  squared again ... only MP1 and 3" — the given values are **already squared**; do not square them again.

### Every t-test in the papers

| Paper          | Context                                    | t           | Critical | Decision             |
|----------------|--------------------------------------------|-------------|----------|----------------------|
| Oct 2020       | thistle leaves, insecticide vs water       | <b>10.3</b> | 2.048    | reject $H_0$         |
| Jan 2021       | E. coli zone of inhibition, village A vs B | <b>2.76</b> | 2.05     | reject $H_0$         |
| Jan 2022       | mayfly nymphs, stream A vs B               | <b>1.03</b> | 2.05     | accept $H_0$         |
| June 2022      | sunflower stem diameter, control vs bent   | <b>7.75</b> | —        | reject $H_0$         |
| Jan 2023       | <i>Nereis</i> extension time, mud vs sand  | <b>1.37</b> | 2.09     | accept $H_0$         |
| June 2023      | pupae, pyrethrum-treated vs untreated      | <b>4.11</b> | 2.14     | reject $H_0$         |
| Oct 2023       | locusts, leaf mass eaten, variety A vs B   | <b>4.32</b> | 2.23     | reject $H_0$         |
| Jan 2024       | lizard toepads, before vs after hurricane  | <b>1.96</b> | 2.12     | accept $H_0$         |
| Oct 2024       | pea mitotic index, 5% vs 20% $O_2$         | <b>11.3</b> | 2.12     | reject $H_0$         |
| Oct 2025       | biomass, 2 cows vs 4 cows                  | <b>2.53</b> | 2.31     | reject $H_0$         |
| Jan 2026       | dung beetle species, site A vs B           | <b>3.50</b> | 2.10     | reject $H_0$         |
| June 2026 (01) | $O_2$ consumption, trained vs untrained    | <b>0.60</b> | 2.31     | accept $H_0$ (model) |

## 30. Spearman's rank correlation

### The formula

$$r_s = 1 - (6 \sum d^2) \div (n(n^2 - 1))$$

1. Rank each variable separately (1 = smallest). **Tied values share the mean rank** (two values tying for 3rd and 4th both get 3.5).

2.  $d$  = difference between the two ranks for each pair; square it; add  $\rightarrow \Sigma d^2$ .
3. Substitute.

### Worked example — Oct 2021 Q3(c), marram grass

$n = 7$ ,  $\Sigma d^2 = 8$ :  $r_s = 1 - (6 \times 8) \div (7 \times 48) = 1 - 48/336 = 1 - 0.143 = \mathbf{0.857}$ . Critical value ( $n = 7$ ,  $p = 0.05$ ) = **0.786**.  $0.857 > 0.786 \rightarrow$  **reject  $H_0$** : there is a significant **positive** correlation between position on the dune and mean leaf length.

### The mark scheme structure

1. **intermediate step** —  $\Sigma d^2$  (or  $6\Sigma d^2 \div n(n^2 - 1) = 0.143$ );
2.  **$r_s$**  — to the right rounding ("*must be correct rounding*" — Oct 2022: 0.929 / 0.93 / 0.9).

### Every Spearman in the papers

| Paper           | Context                                | $r_s$                                                     | Critical     | Decision                             |
|-----------------|----------------------------------------|-----------------------------------------------------------|--------------|--------------------------------------|
| Oct 2021        | marram leaf length vs dune position    | <b>0.857</b>                                              | 0.786        | reject — positive correlation        |
| Oct 2022        | % germination vs arsenic concentration | <b>0.929</b> ( $\Sigma d^2 = 4$ , so $6\Sigma d^2 = 24$ ) | 0.786        | reject — correlation                 |
| Jan 2025        | % gill visible vs number of touches    | <b>0.1</b> ( $\Sigma d^2 = 18$ , $n = 5$ )                | <b>1.000</b> | accept — no correlation              |
| June 2025       | % oil remaining vs days stored         | <b>-0.857</b> ( $\Sigma d^2 = 104$ )                      | 0.786        | reject — <b>negative</b> correlation |
| June 2026 (01A) | tidal volume vs height                 | <b>0.714</b> ( $6\Sigma d^2 = 96$ )                       | 0.786        | accept — no correlation              |

**The sign:** June 2025 required the minus sign in the answer ("*must have – sign*"), but in the comparison "*Ignore – sign on any calculated value*": compare **0.857** with 0.786. Say whether the correlation is **positive** or **negative**.

**$n = 5$ :** the critical value at  $p = 0.05$  is **1.000** — only a perfect correlation is significant. This is itself a limitation: **too few data points** — use more values of the IV.

## 31. Chi-squared

### The formula

$$\chi^2 = \sum (O - E)^2 \div E; \text{ degrees of freedom} = \text{number of categories} - 1$$

**Expected values:** if there is no preference / no difference,  $E = \text{total} \div \text{number of categories}$ .

### Worked example — June 2024 Q3(b), topshell habitat preference

| Condition     | O  | E  | $(O - E)^2 \div E$  |
|---------------|----|----|---------------------|
| smooth, light | 12 | 20 | $64 \div 20 = 3.20$ |
| rough, light  | 15 | 20 | $25 \div 20 = 1.25$ |
| smooth, dark  | 24 | 20 | $16 \div 20 = 0.80$ |
| rough, dark   | 29 | 20 | $81 \div 20 = 4.05$ |
|               | 80 | 80 | $\chi^2 = 9.30$     |

$df = 4 - 1 = 3 \rightarrow$  critical value **7.815**.  $9.3 > 7.815 \rightarrow$  **reject  $H_0$** : the topshells **show a preference** — they **prefer dark** to light; little difference between rough and smooth.

### Every $\chi^2$ in the papers

| Paper     | Context                                                                     | $\chi^2$    | Critical | Decision                     |
|-----------|-----------------------------------------------------------------------------|-------------|----------|------------------------------|
| June 2022 | zebrafish colour choice ( $E = 36$ ; $O = 44, 28$ )                         | <b>3.56</b> | 3.84     | accept — no preference       |
| June 2024 | topshell habitats                                                           | <b>9.3</b>  | 7.815    | reject — preference for dark |
| Jan 2025  | fruiting acacias near / far from termite mounds ( $E = 65$ ; $O = 79, 51$ ) | <b>6.03</b> | 3.84     | reject — difference          |

*June 2022 ruling: "Must state the calculated / chi squared value is more / less than the critical value / 3.84." A decision without the comparison scores nothing.*

## 32. Mann–Whitney U

The formula (as printed in *June 2021*)

$$U_1 = R_1 - n_1(n_1 + 1) \div 2 \quad U_2 = R_2 - n_2(n_2 + 1) \div 2$$

R = the **sum of ranks** of that group when **all** the data from both groups are ranked together.

**Decision rule** — the opposite way round from the other tests: the difference is significant if the **smallest U is equal to or less than** the critical value.

### Worked example — *June 2021 Q2(d), switchgrass*

$n_1 = n_2 = 7$ ;  $R_1 = 57.5$ ,  $R_2 = 47.5$  (check:  $57.5 + 47.5 = 105 = 1 + 2 + \dots + 14 \checkmark$ ).  $U_1 = 57.5 - 28 = 29.5$ ;  $U_2 = 47.5 - 28 = \mathbf{19.5}$ . Smallest U = 19.5; critical value ( $p = 0.05$ ) = **8**.  $19.5 > 8 \rightarrow$  **accept  $H_0$** : there is **no significant difference** in biomass between Almo and Kanlow; the range bars **overlap**.

*Oct 2019*:  $U = 18.0$ , critical value 30  $\rightarrow 18 < 30 \rightarrow$  **reject  $H_0$** : a significant difference in shell length between snails on diets A and B.

### When to use it

Data **not normally distributed**, small samples, or **ordinal** data. It compares **medians**, so the table shows **ranked data** and the **median**, and the bar graph plots **medians**.

---

## 33. Other calculations

### 33.1 Percentage change

$$\% \text{ change} = (\text{new} - \text{original}) \div \text{original} \times 100$$

*June 2022*:  $(223.3 - 92.7) \div 92.7 \times 100 = \mathbf{141\%}$  (to 3 s.f.). *Oct 2024*: kenaf **10.4%**. *June 2024*: zebrafish distance decrease **71%** ("Ignore minus sign"). *Oct 2020*: 26 / 26.1%. *Oct 2023*: 23%.

### 33.2 Magnification and actual size

$$\text{magnification} = \text{image size} \div \text{actual size}; \text{actual size} = \text{image size} \div \text{magnification}$$

Convert to the **same units** first (1 cm = 10 mm = 10 000  $\mu\text{m}$ ). *June 2024*: topshells  $\times 2.5$  from a 1 cm scale bar. *June 2025*: earthworm image **smaller than life**  $\rightarrow$  magnification **0.37–0.5**. *June 2026 (01)*: beetle at  $\times 15 \rightarrow$  length in **mm**.

### 33.3 Rates

rate = change ÷ time, with **units** (a separate mark)

Pollen tube **20.0  $\mu\text{m h}^{-1}$** ; maggot growth **0.33 mm h<sup>-1</sup>**; yeast turbidity **1.6 a.u. h<sup>-1</sup>**; O<sub>2</sub> uptake **6.1–6.5 × 10<sup>-3</sup> dm<sup>3</sup> s<sup>-1</sup>**. "Allow /h / hr / hour" for the unit in Jan 2024, but "Not dm<sup>3</sup>/s" in Oct 2023 — write **h<sup>-1</sup> / s<sup>-1</sup> / per hour** to be safe.

### 33.4 Volumes

- **Gas in a capillary:**  $V = \pi r^2 \times \text{distance}$  (Jan 2021: 0.14 mm<sup>3</sup> min<sup>-1</sup>). Radius = **diameter ÷ 2**.
- **Sphere:**  $V = \frac{4}{3} \pi r^3$  (Jan 2022: drops of diameter 3.0 mm,  $r = 1.5 \rightarrow 14.1 \text{ mm}^3$  each, two drops a minute  $\rightarrow 28 \text{ mm}^3$  to 2 s.f.).
- **Dilution** (Oct 2022): 200 mg dm<sup>-3</sup>  $\rightarrow$  **4 mg in 20 cm<sup>3</sup>** (0.004 g in 20 cm<sup>3</sup>).

### 33.5 Respiratory quotient

**RQ = CO<sub>2</sub> produced ÷ O<sub>2</sub> used** — with soda lime ( $x = \text{O}_2$ ) and without ( $y = \text{O}_2 - \text{CO}_2$ ): **RQ =  $(x - y) \div x$**

Jan 2021:  $0.11 \div 0.14 = \mathbf{0.79}$ . Carbohydrate  $\approx 1.0$ ; lipid  $\approx 0.7$ ; protein  $\approx 0.9$ .

### 33.6 Spirometer

**Minute ventilation = tidal volume × breathing rate**; O<sub>2</sub> uptake = gradient of the troughs (§12).

### 33.7 Standard deviation

$$s = \sqrt{(\Sigma(x - \bar{x})^2 \div (n - 1))}$$

June 2025 Q2(b)(ii):  $\Sigma(x - \bar{x})^2 = 19.39$ ;  $\div 4 = 4.85$ ;  $\sqrt{\phantom{x}} = \mathbf{2.2}$ . The scheme gave MP1 for any of the intermediate values 10.24 / 19.39 / 4.85 — **show them**.

Use of SDs (June 2025 (b)(iii)): **compare the means**; then use the **SDs** to see if the difference is **significant** — by a **t-test** or by checking whether **mean  $\pm$  SD overlap**.

### 33.8 Index of diversity

$$D = N(N - 1) \div \Sigma n(n - 1)$$

June 2026 (01) Q2 — model answer:  $N = 12 + 18 + 1 + 4 + 6 = 41$ ;  $N(N - 1) = 41 \times 40 = 1640$ ;  $\Sigma n(n - 1) = 132 + 306 + 0 + 12 + 30 = 480$ ; **D =  $1640 \div 480 = 3.42$**  (2 d.p.).

Sustainability ((b)(iii), model): D rises from 3.42 (3 years) to 3.58 (10 years) to **5.15 after 20 years** — **the same as forest never cleared (5.15)**, so the forest **fully recovers its biodiversity** if left for 20

years before re-clearing.

### 33.9 Odds ratio

June 2023 Q2(c): Step 1 =  $60 \div 16 = 3.75$ ; Step 2 =  $4 \div 20 = 0.2$ ; **odds ratio =  $3.75 \div 0.2 = 18.75 \rightarrow 19$**  (2 s.f.; "not 19.0").  $> 1 \rightarrow$  the presence of mimosa trees **increases the likelihood** of cactus plants; conclusion: "The cactus plants are found growing in association with the mimosa".

### 33.10 Growth-rate constant (bacteria)

$$k = (\log_{10} N_t - \log_{10} N_0) \div (\log_{10} 2 \times t) \text{ — Jan 2023: } \mathbf{1.99} \text{ (see §14.4).}$$

### 33.11 $Q_{10}$ (temperature coefficient)

$$Q_{10} = \text{rate at } (T + 10 \text{ }^{\circ}\text{C}) \div \text{rate at } T$$

June 2026 01A:  $17.6 \text{ bubbles min}^{-1} \times Q_{10} \text{ of } 2 = \mathbf{35.2}$ . Enzyme-controlled reactions have  $Q_{10} \approx 2\text{--}3$  between 0 and  $40 \text{ }^{\circ}\text{C}$ .

### 33.12 Energy from biomass

June 2021 Q2(b)(ii): **162 450 MJ  $\rightarrow 1.6 \times 10^5 \text{ MJ}$**  — "expressed to two significant figures in standard form". **Net primary productivity / biomass** definition ((b)(i)): "the organic matter of living organisms in a given area" (per unit time).

### 33.13 Significant figures — the rules the schemes enforce

| Paper         | Demand                                                                    |
|---------------|---------------------------------------------------------------------------|
| Jan 2022      | $28 \text{ mm}^3$ — <b>2 s.f.</b> is a mark                               |
| June 2022     | 141 — "Answer must be to 3 sig figs"                                      |
| Jan 2023      | $k = 1.99$ — "Allow 2 marks for a correct answer with the wrong sig figs" |
| June 2023     | 19 — "not 19.0"                                                           |
| June 2021     | $1.6 \times 10^5$ — 2 s.f. <b>in standard form</b>                        |
| Jan 2026      | 1.6 — 2 s.f.                                                              |
| June 2026 01A | 0.14–0.16 — "answer given to two significant figures" is a separate mark  |

**Always** re-read the question for "to two significant figures" / "to two decimal places" and underline it.

## 34. Writing the conclusion from a statistical test

### The three- or four-point answer

1. **State the critical value and compare:** "The calculated value of  $t$  (**4.32**) is **greater than** the critical value (**2.23**) at  $p = 0.05$ ."
2. **Decision:** "therefore reject the null hypothesis."
3. **In context:** "so there is a significant difference in the mass of leaves eaten between variety A and variety B" — and, where possible, **which way:** "locusts ate **more** of variety B."
4. **Comment on variability:** "the **range bars do not overlap**" (or overlap), which supports / weakens the conclusion.

### Rulings

| Ruling                                                                                                                    | Paper                                             |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| "Accept critical value circled / indicated in table"                                                                      | Oct 2022, June 2021, Oct 2025                     |
| "Do not allow –ve values / $t < \text{critical value}$ if $t$ value not given" — <b>quote the number</b>                  | Oct 2023                                          |
| "Reject other critical values" — use the right df / $n$ / $p = 0.05$                                                      | June 2025                                         |
| directional first point: "under 20% oxygen have a larger mitotic index"; "a field with 4 cows has a lower (mean) biomass" | Oct 2024, Oct 2025                                |
| "Do not accept reference to field C / 0 cows and significance" — you only <b>tested</b> A vs B                            | Oct 2025                                          |
| "comment on variability of data — eg range bars overlap"; "IGNORE reliability / accuracy / repeatability"                 | Oct 2020, June 2021, Jan 2023, Oct 2022, Jan 2026 |
| "Ignore any comment on variability" on a Spearman in Oct 2021 — harmless, but not needed there                            | Oct 2021                                          |
| $\chi^2$ : "top shells show a preference" + <b>which</b> preference                                                       | June 2024                                         |
| ECF throughout — a wrong $t$ , correctly compared, still earns the decision marks                                         | all                                               |

### The decision table

| Test     | Significant (reject $H_0$ ) when... |
|----------|-------------------------------------|
| t-test   | $t \geq \text{critical}$            |
| Spearman | **                                  |

| Test         | Significant (reject $H_0$ ) when...                            |
|--------------|----------------------------------------------------------------|
| $\chi^2$     | $\chi^2 \geq \text{critical}$                                  |
| Mann–Whitney | <b>smallest U <math>\leq</math> critical</b> ← the odd one out |

"Accept the null hypothesis" is the phrase the schemes use for a non-significant result. Also acceptable: "the difference is **not significant** / is due to **chance**".

## 35. Degrees of freedom and critical values — quick reference

| Test            | df / n  | Typical critical value ( $p = 0.05$ ) seen in the papers |
|-----------------|---------|----------------------------------------------------------|
| t-test, 5 + 5   | df 8    | 2.31                                                     |
| t-test, 6 + 6   | df 10   | 2.23                                                     |
| t-test, 8 + 8   | df 14   | 2.14                                                     |
| t-test, 9 + 9   | df 16   | 2.12                                                     |
| t-test, 10 + 10 | df 18   | 2.10                                                     |
| t-test, 11 + 11 | df 20   | 2.09                                                     |
| t-test, 15 + 15 | df 28   | 2.05                                                     |
| Spearman        | n = 7   | 0.786                                                    |
| Spearman        | n = 5   | 1.000                                                    |
| $\chi^2$        | df 1    | 3.84                                                     |
| $\chi^2$        | df 3    | 7.815                                                    |
| Mann–Whitney    | 7 + 7   | 8                                                        |
| Mann–Whitney    | 11 + 11 | 30                                                       |

**p = 0.05** means there is a **5% probability** that a difference / correlation this large is due to **chance**. Always use the **p = 0.05** column unless told otherwise (some schemes also accepted the  $p = 0.01$  value with correct reasoning).

## PART E — VALIDITY, IMPROVEMENTS AND EXTENSIONS

### 36. "Explain why this conclusion may not be valid" / "Comment on the validity"

These 2–4 mark questions follow the data question. The schemes reward **specific** criticisms of **this** investigation. Generic "repeat it" / "small sample" / "not reliable" is usually ignored.

#### 36.1 The six families of validity point

| Family                                 | Model point                                                                                                                                                                        | Seen in                                        |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| 1. Variability of the data             | "the <b>range bars overlap</b> , so the difference may be due to chance"                                                                                                           | <i>Oct 2019, Jan 2022, June 2022</i>           |
| 2. Uncontrolled / unmeasured variables | " <b>named</b> environmental variable (e.g. <b>soil moisture / shading / temperature / pH / flow rate / light / other pollutants</b> ) was not controlled or measured"             | <i>June 2021, Oct 2019, Jan 2022, Oct 2020</i> |
| 3. Sampling                            | "only <b>one field / two streams</b> — may not be <b>representative</b> "; "plots may not have been <b>randomly selected</b> — e.g. all Almo plots on one side of the field"       | <i>June 2021, Jan 2022</i>                     |
| 4. Organisms not matched               | "plants may be <b>different varieties / genetically different</b> "; " <b>source</b> of snails / genetic variability"                                                              | <i>June 2022, Oct 2019</i>                     |
| 5. Time and baseline                   | "leaves should have been <b>measured before the treatment</b> "; "samples on <b>more than one day / time of day</b> "; "range over <b>seven years</b> — weather differs each year" | <i>Oct 2020, Jan 2022, June 2021</i>           |
| 6. Alternative explanation             | "plants may bend for <b>other reasons, e.g. phototropism</b> "; "pyrethrum may be <b>washed off</b> fruit / <b>not reach the eggs</b> / flies may be <b>resistant</b> "            | <i>June 2022, June 2023</i>                    |

#### 36.2 The rulings

| Ruling                                                        | Paper            |
|---------------------------------------------------------------|------------------|
| "Do not accept small sample size"                             | <i>June 2021</i> |
| "IGNORE different fields"                                     | <i>Oct 2020</i>  |
| "IGNORE age / initial length / gender" for the snail variable | <i>Oct 2019</i>  |

| Ruling                                                                                            | Paper     |
|---------------------------------------------------------------------------------------------------|-----------|
| "Ignore IAA / auxins" — the mark is for an <b>alternative cause</b> of bending, not the mechanism | June 2022 |
| "Accept the investigation was not repeated" (Oct 2019 only — do not rely on it)                   | Oct 2019  |

### 36.3 A model 3-mark answer (June 2021 Q2(e), switchgrass)

"Only **one field** was used, so the results may not be **representative** of these varieties. The plots of each variety may not have been **randomly placed** — the Almo plots could all be on one side of the field. A named condition such as **soil moisture or shading** may not have been the same for both varieties. Also, **weather conditions varied over the seven years**, which is why the range of data is so large."

## 37. "Suggest how this investigation could be extended / improved"

### 37.1 Extension — change one thing and investigate further

The schemes: "ignore repeat the expt / amount" (June 2023); "Ignore repeat the expt / use more locusts" (Oct 2023); "Ignore increase sample size / repeat the experiment" (Jan 2025); "Ignore repeat unqualified / bigger fields" (Oct 2025).

**The template: a new IV value / a new variable / a longer time / a new species or place.**

| Paper                    | Accepted extensions                                                                                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| June 2023<br>(pyrethrum) | <b>other masses / concentrations</b> of pyrethrum; leave adults to lay eggs > <b>24 h</b> ; leave eggs > <b>5 days</b>                                                                                       |
| Oct 2023<br>(locusts)    | <b>different temperatures / humidities</b> ; <b>different species / ages / stages</b> of locust; <b>longer than 24 h</b> ; offer <b>both varieties at once</b> so locusts can choose                         |
| Jan 2024<br>(lizards)    | <b>repeat after each hurricane</b> ; measure <b>hind-limb</b> toepads; <b>other locations</b> ; <b>other species</b>                                                                                         |
| Oct 2024 (mitotic index) | <b>different temperatures</b> ; <b>other O<sub>2</sub> concentrations</b> (within or beyond 5–20%); <b>different times</b> ; <b>other species</b>                                                            |
| Jan 2025<br>(Aplysia)    | <b>more than 5 touches</b> ; <b>change the interval between touches</b> ; <b>more sea slugs / different ages / species</b> ; change a <b>sea-water variable</b> (temperature, pH, salinity, O <sub>2</sub> ) |
| Oct 2025 (cows)          | <b>change the number / type</b> of cows (more values → correlation test); <b>more biomass samples</b> ; <b>different times of year</b> ; <b>different grass species / more fields</b>                        |

| Paper                   | Accepted extensions                                                                                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jan 2026 (dung beetles) | <b>other locations / other % grass cover; more than 10 pitfall traps; traps for &gt; 5 days; different times of year</b>                                                       |
| Jan 2024 (pollen)       | <b>store pollen for different times</b> before use; keep a named variable the same                                                                                             |
| Oct 2022 (arsenic)      | germinate with <b>no arsenic</b> as well as <b>different concentrations</b> ; control a stated condition; measure <b>height / mass</b> after a <b>stated time</b> (7–100 days) |

### 37.2 Improvement — make the same investigation more valid

| Paper                                        | Accepted improvements                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Oct 2021 (marram)                            | <b>more plants at each position (&gt; 10) / another transect; more positions</b> on the dune (smaller intervals); <b>measuring tape instead of a ruler</b> (leaves > 1 m); <b>avoid bias — random sampling</b> / always the longest leaf                                                                                                |
| June 2024 (topshells) — each with its reason | <b>leave them longer</b> <i>so they have time to select a habitat</i> ; <b>more than five</b> molluscs <i>for more data</i> ; <b>control temperature</b> <i>so they move at the same speed</i> ; <b>cover the light side with glass</b> <i>so both sides are covered</i> ; a <b>round tray</b> <i>so they do not collect in corners</i> |
| Oct 2020 (thistle)                           | measure leaves <b>before</b> treatment; <b>control / measure</b> environmental conditions (greenhouse); measure at <b>different time intervals</b> ; <b>different insecticide concentrations</b>                                                                                                                                        |

**Improvement answers score best with a reason:** "...so that / because...". June 2024 gave the reason inside each marking point.

### 37.3 Limitations in the planning question — link back

The planning limitations bank is in §5. The June 2026 01A anaerobic-yeast scheme is a good model: *"uneven distribution of yeast / sugar"; "difficult to judge the end point (redox indicator)" / "difficult to measure volume of gas"; "aerobic respiration may occur if yeast is not left to use the oxygen" — "DO NOT ACCEPT contamination", "IGNORE pH".*

Oct 2020 (Cabomba plan): *"difficult to control all variables affecting photosynthesis"; a named hard-to-control variable; leaf surface area cannot be controlled / measured; errors measuring the volume of gas — "size of bubble may vary", "Not counting bubbles unqualified".*

## PART F — THE THEORY MARKS INSIDE THE PRACTICAL PAPER

### 38. Every theory question, grouped by topic

Each practical paper carries **5–12 marks of straight biology** tied to the context. They come from Units 1–5. The model answers below are the mark-scheme points.

#### 38.1 Biological molecules

| Question                                  | Mark points                                                                                                                                              | Paper               |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>starch</b> — why suited to storage (2) | polymer of <b>glucose</b> ; <b>large</b> → <b>insoluble / no osmotic effect</b> ; <b>branched</b> → <b>hydrolysed quickly</b> (" <i>IGNORE easily</i> ") | Oct 2020            |
| <b>cellulose</b> structure (3)            | <b>β-glucose</b> ; <b>1–4 glycosidic bonds</b> ; <b>every other glucose inverted</b> ; <b>straight / unbranched chain</b>                                | Jan 2022, June 2023 |
| <b>collagen</b> structure (3)             | <b>polypeptide</b> ; <b>triple helix</b> ; <b>hydrogen bonds</b> ; <b>fibrous</b> ; <b>glycine every third</b> (" <i>Not β-pleated sheet</i> ")          | Oct 2023            |
| <b>maltose</b> (2)                        | disaccharide of <b>two α-glucose</b> (" <i>Do not accept beta</i> "); <b>1–4 glycosidic bond</b> (" <i>Do not accept 1–6</i> ")                          | Oct 2025            |
| <b>pectin / tube-tip growth</b> (4)       | starch <b>hydrolysed</b> ; <b>condensation</b> → pectin; <b>vesicles</b> ; <b>fuse with membrane / exocytosis</b> ; <b>ATP from mitochondria</b>         | Jan 2021            |

#### 38.2 Enzymes

| Question                       | Mark points                                                                                                                                                                         | Paper                              |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| temperature and rate (3–4)     | <b>kinetic energy</b> ; <b>more collisions per unit time / more ES complexes</b> ; enzymes (e.g. <b>RUBISCO</b> ); above optimum <b>denature</b> → <b>active site changes shape</b> | June 2024, Jan 2024, June 2026 01A |
| <b>calcium as cofactor</b> (3) | <b>substrate binds to active site</b> ; <b>lowers activation energy</b> ; faster <b>respiration</b> ; faster <b>anabolic</b> reactions                                              | June 2021                          |
| <b>copper as inhibitor</b> (3) | interferes with <b>ionic bonds / R groups</b> ; alters <b>tertiary structure</b> ; <b>active site shape changed</b> ; <b>fewer ES complexes</b>                                     | Jan 2025                           |
| <b>gibberellin</b> (3)         | <b>transcription factor</b> ; <b>transcription of amylase mRNA</b> ; <b>translation of amylase</b>                                                                                  | Oct 2024                           |

### 38.3 Respiration and photosynthesis

| Question                                                               | Mark points                                                                                                                                                 | Paper                 |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| role of oxygen (3)                                                     | needed for <b>aerobic</b> respiration; <b>final electron acceptor</b> in <b>oxidative phosphorylation</b> ; low O <sub>2</sub> → <b>anaerobic</b>           | Oct 2019              |
| where redox indicator acts (2)                                         | <b>cristae / inner membrane</b> ; normal carriers <b>NAD / FAD</b>                                                                                          | June 2024             |
| ATP in anaerobic respiration (3)                                       | co-enzyme <b>NAD</b> ; electrons to NAD; ADP + Pi → ATP by <b>substrate-level phosphorylation</b> (" <i>not oxidative</i> ")                                | June 2025             |
| <b>lactate after exercise</b> (2)<br>( <i>model — no mark scheme</i> ) | lactate → <b>liver</b> ; <b>oxidised back to pyruvate</b> (using O <sub>2</sub> — the oxygen debt), then <b>converted to glucose / glycogen</b> or respired | June 2026<br>(01)     |
| <b>grana</b> (3)                                                       | <b>stacked membranes</b> → <b>large surface area</b> ; <b>pigments</b> ; <b>electron carriers / enzymes</b>                                                 | Oct 2021              |
| <b>photolysis</b> (3)                                                  | water split; <b>H<sub>2</sub>O → 2H<sup>+</sup> + 2e<sup>-</sup> + ½O<sub>2</sub></b> ; electrons removed; O atoms → O <sub>2</sub>                         | Oct 2022              |
| <b>nitrate</b> in plants (2–3)                                         | <b>amino acids / proteins / enzymes (RUBISCO)</b> ; <b>nucleotides / DNA / RNA</b> → <b>cell division</b> ; <b>chlorophyll</b>                              | Jan 2023,<br>Jan 2025 |

### 38.4 Transport, exchange and ventilation

| Question                                | Mark points                                                                                                                                                                                                                                                                                         | Paper    |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| why breathing increases in exercise (2) | more O <sub>2</sub> for <b>aerobic respiration</b> ; muscles need <b>ATP</b> ; more <b>CO<sub>2</sub></b> to remove; <b>oxidise lactate</b>                                                                                                                                                         | Oct 2021 |
| <b>control of ventilation</b> (3)       | <b>chemoreceptors</b> detect ↑CO <sub>2</sub> / ↓pH; <b>impulses</b> to <b>intercostal muscles / diaphragm</b> ; rate increases                                                                                                                                                                     | Oct 2023 |
| <b>caffeine uptake by skin</b> (2)      | <b>facilitated diffusion</b> via <b>channel proteins</b> ; <b>concentration gradient falls</b> over time                                                                                                                                                                                            | Oct 2022 |
| <b>xylem vs phloem</b> (4)              | xylem: <b>larger lumen</b> , <b>dead / hollow</b> , <b>no end walls</b> , <b>lignified</b> , <b>water + ions</b> , <b>one way</b> , <b>support</b> , <b>passive</b> ; phloem: <b>living / companion cells</b> , <b>sieve plates</b> , <b>sucrose / amino acids</b> , <b>two way</b> , <b>active</b> | Jan 2026 |
| <b>sclerenchyma</b> (2)                 | <b>lignin / secondary thickening</b> ; <b>end walls</b> ; <b>dead / hollow</b> ; <b>pits</b>                                                                                                                                                                                                        | Oct 2024 |

| Question                                           | Mark points                                                                                                     | Paper     |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------|
| <b>blood clotting</b> (4, in order)                | <b>thromboplastin; prothrombin → thrombin; fibrinogen → fibrin; traps RBCs / platelets</b>                      | June 2022 |
| <b>water potential differs between species</b> (3) | different <b>solutes / concentrations</b> → different <b>ψ</b> ; <b>cell walls / turgor</b> differ              | Oct 2020  |
| water and the snail membrane (2)                   | <b>presence</b> of water, not volume, matters; breakdown <b>limited by another factor</b> (enzymes / thickness) | Jan 2022  |

### 38.5 Microbiology, immunity and evolution

| Question                                             | Mark points                                                                                                                                                              | Paper                    |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>bacteriostatic vs bactericidal</b> (1–2)          | <b>prevents increase in numbers vs kills bacteria</b>                                                                                                                    | Jan 2024, June 2026 (01) |
| <b>prescribing code</b> (2)                          | only for <b>bacterial</b> infections; <b>finish the course</b> → reduces <b>resistance</b> / selection pressure                                                          | Jan 2024                 |
| <b>insecticide resistance</b> (3) ( <i>model</i> )   | <b>mutation; selection pressure</b> ; resistant <b>survive and reproduce</b> ; <b>allele frequency</b> rises                                                             | June 2026 (01)           |
| <b>seed varieties differ in arsenic response</b> (2) | differences in <b>alleles / DNA</b> ; some varieties <b>cannot absorb</b> arsenic / membrane permeability; <b>mutation / natural selection</b> (" <i>Ignore genes</i> ") | Oct 2022                 |
| <b>biofuels</b> (1)                                  | release only CO <sub>2</sub> <b>absorbed in photosynthesis</b> — <b>carbon neutral</b>                                                                                   | Jan 2022                 |

### 38.6 Behaviour, nerves and ecology

| Question                                   | Mark points                                                                                                                                                                   | Paper              |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>opsin / reflex</b> in <i>Nereis</i> (2) | light changes <b>retinal cis</b> → <b>trans</b> / opsin changes; <b>action potential</b> in sensory neurone; <b>reflex</b> / muscle contraction (" <i>Ignore rhodopsin</i> ") | Jan 2023           |
| <b>advantage of habituation</b> (1–3)      | <b>saves energy</b> ; allows <b>feeding / reproduction</b> ; <b>avoids damage</b> ; <i>Aplysia</i> : <b>more O<sub>2</sub> uptake</b>                                         | Oct 2025, Jan 2025 |
| <b>mimosa and cactus</b> (1)               | <b>shelter</b> from wind / sun → <b>less transpiration</b> ; seedlings <b>establish</b>                                                                                       | June 2023          |

| Question                       | Mark points                                                                                              | Paper     |
|--------------------------------|----------------------------------------------------------------------------------------------------------|-----------|
| climate change and locusts (1) | locusts <b>eat more / respire faster / grow faster</b> at higher temperature; complete life cycle faster | Oct 2023  |
| biomass / NPP (1)              | <b>organic matter</b> of living organisms <b>in a given area</b> (per unit time)                         | June 2021 |

**Theory marks are Unit 1–5 content wearing a practical coat.** The chains in §38 are the same chains as in the Unit 4 and Unit 5 notes — revise them there, and practise writing them in this exact mark-point language.

## PART G — EXAM TECHNIQUE

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### 39. Command words in Unit 6

| Command word                          | What the scheme wants                                                                                                 | Example                                               |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>State / Give / Identify / Name</b> | one word or phrase; <b>first answer on each line is marked</b>                                                        | "State two abiotic variables"                         |
| <b>Describe</b> (a method)            | a sequence of <b>method points</b> , each <b>specific</b> : values, apparatus, times                                  | "Describe how the student could investigate..." (4–6) |
| <b>Describe</b> (data)                | trend + <b>figures with units</b>                                                                                     | "Describe the conclusion..."                          |
| <b>Explain</b>                        | a point <b>plus a reason</b> — <i>because / so that</i>                                                               | "Explain why sodium hydrogen carbonate was added"     |
| <b>Calculate</b>                      | <b>working</b> → <b>answer</b> → <b>unit</b> → <b>s.f.</b>                                                            | t, r <sub>s</sub> , $\chi^2$ , rate, % change         |
| <b>Suggest</b>                        | apply knowledge to an unfamiliar context — any <b>valid</b> answer scores (" <i>Accept other valid suggestions</i> ") | "Suggest one benefit to the cactus"                   |
| <b>Plan / Devise</b>                  | the full plan under the headings (§1–§7)                                                                              | Q4                                                    |
| <b>Comment on / Evaluate</b>          | validity: variability, variables, sampling (§36)                                                                      | "Comment on the validity of the conclusion"           |
| <b>Draw</b> (table / graph)           | the 3-mark formats in §27–§28                                                                                         |                                                       |

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### 40. Phrase bank — exact wording that scores

#### Method

- "The **dependent variable** is the {**volume of oxygen / time for the indicator to change colour / number of eggs hatched**} measured **per {minute / stated time}**."
- "Use **five** {concentrations / temperatures}: **0, 0.25, 0.50, 0.75 and 1.00 mol dm<sup>-3</sup>**."
- "**Control temperature** with a **thermostatic water bath** at **25 °C**."
- "**Control pH** with a **buffer** at **pH 7**."

- "Keep **light intensity** constant using the **same wattage bulb** at a **fixed distance** of **20 cm**, with a **heat shield**."
- "Use the **same mass** of seeds, measured on a **balance**."
- "Leave to **acclimatise / equilibrate** for **10 minutes** before starting."
- "**Repeat** at least **three / five times** at each {concentration} and **calculate a mean** (and **standard deviation**)."
- "**Rate = volume ÷ time**."

### Preliminary work

- "**Find a suitable** {range of concentrations / mass of seeds / time} **to give measurable results**."
- "**Find a suitable** method to **measure** {the DV}."

### Recording, presenting and analysing

- "Record the raw data in a **table** with the IV in the first column and repeats and **mean** in the other columns, **units in the headings**."
- "Plot a {**bar graph of the means with range bars / scatter graph of the IV against the mean DV**}."
- "Use a {**Student's t-test / Spearman's rank correlation**} to see whether the {**difference / correlation**} is **significant at  $p = 0.05$** ."

### Variables

- "If temperature were **higher**, **enzyme activity would increase**, so **more** CO<sub>2</sub> would be produced — the results would **not be valid**."

### Conclusions

- "The calculated value of  $t$  (**4.11**) is **greater than** the critical value (**2.14**) at  $p = 0.05$ , **so reject the null hypothesis**: there is a **significant difference** in the number of pupae between the treated and untreated media. The **range bars do not overlap**."
- "The calculated  $r_s$  (**0.714**) is **less than** the critical value (**0.786**), **so accept the null hypothesis**: there is **no significant correlation** between height and tidal volume."

### Validity

- "**Only one field** was used, so the sample **may not be representative**."
- "A **named variable** such as **soil moisture** was **not controlled / measured**."
- "The **range bars overlap**, so the difference **may be due to chance**."

## 41. The 25 most lost marks in Unit 6

1. **"Amount"** instead of volume / mass / concentration / number — *ignored* in every scheme.
  2. **"Different temperatures"** with no values — state **five** and the **range**.
  3. **A controlled variable without the method** — "keep temperature constant" is half a mark at best; add **"thermostatic water bath"**.
  4. **"Water bath"** unqualified — *"Ignore waterbath unqualified"*: say **thermostatic**.
  5. **"Rate of photosynthesis"** as the DV — give the **measured quantity**: volume of O<sub>2</sub> per minute.
  6. **Non-directional effect** — "it would affect the results" = 0.
  7. **"Not reliable / not accurate"** as the effect — *ignored*; write **"not valid"** or a directional effect.
  8. **Null hypothesis with "correlation" for a difference test** (or vice versa) — **"DO NOT ACCEPT"**.
  9. **Null hypothesis with unqualified groups** — "sample A and B" → **"before and after the hurricane"**.
  10. **Units in the body of the table** — *"Do not accept"*.
  11. **An extra column** (trial number, difference) — *"A third column negates MP1"*.
  12. **Means to a different number of decimal places** from the data — *"Not 17 / 18"*.
  13. **Bar graph axis not starting at zero / broken** — the scale mark is lost.
  14. **Range bars drawn as  $\pm$  SD** or missing — range = **lowest to highest**.
  15. **Extrapolating a line** to the origin or beyond the data.
  16. **Squaring the given (s)<sup>2</sup> values again** in the t-test.
  17. **Decision without the comparison** — *"Must state the calculated value is more / less than the critical value"*.
  18. **Mann–Whitney decision the wrong way round** — significant when **U  $\leq$  critical**.
  19. **Forgetting the sign / direction** of a Spearman correlation, or of a t-test difference ("...has a larger mitotic index").
  20. **Missing units on a rate** —  $\mu\text{m h}^{-1}$ ,  $\text{mm h}^{-1}$ ,  $\text{dm}^3 \text{s}^{-1}$  (*"Not dm<sup>3</sup>/s"*).
  21. **Wrong significant figures** — 2 s.f. / 3 s.f. / standard form is a separate mark.
  22. **Using the diameter as the radius** in  $\pi r^2$  or  $\frac{4}{3} \pi r^3$ .
  23. **"Practise the method"** as a preliminary point — *ignored*; write **"find a suitable..."**.
  24. **Risk without a matching reduction** — "wear a lab coat", "be careful", "contamination unqualified".
  25. **"Repeat the experiment" / "bigger sample"** as an extension — *ignored*; change **one named variable** or the **time / place / species**.
-

## 42. Per-paper map

| Series                     | Q1                                                 | Q2                                                              | Q3                                              | Q4 (plan)                           |
|----------------------------|----------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------|-------------------------------------|
| <b>Oct 2019</b><br>(WBI06) | mung bean<br>respirometer,<br>temperature (12)     | snail diets — <b>Mann–Whitney</b> (16)                          | <b>clover</b> + soil<br>water plan, QWC<br>(22) | —                                   |
| <b>Oct 2020</b>            | potato <b>water potential</b> + starch<br>(14)     | thistle insecticide — <b>t</b><br><b>= 10.3</b> (15)            | <b>Cabomba light</b><br>plan (21)               | —                                   |
| <b>Jan 2021</b>            | <b>pollen tube</b> + pectin                        | <b>maggot respirometer</b> , RQ<br>0.79                         | E. coli village A<br>vs B — <b>t = 2.76</b>     | <b>ABA</b> and seeds                |
| <b>June 2021</b>           | <b>brine shrimp</b> calcium                        | switchgrass — <b>Mann–Whitney</b> , U = 19.5                    | <b>Aplysia</b><br>habituation plan              | —                                   |
| <b>Oct 2021</b>            | <b>Elodea wavelength</b> +<br>grana                | <b>spirometer</b> , minute<br>ventilation                       | marram — <b>r_s = 0.857</b>                     | <b>pineapple</b><br>antimicrobial   |
| <b>Jan 2022</b>            | kenaf <b>tensile strength</b> + cellulose          | snail membrane,<br>sphere volume 28<br>mm <sup>3</sup>          | mayfly — <b>t = 1.03</b>                        | <b>yeast temperature</b>            |
| <b>June 2022</b>           | zebrafish — <b><math>\chi^2 = 3.56</math></b>      | <b>vitamin C / DCPIP</b> +<br>clotting                          | sunflower — <b>t = 7.75</b>                     | <b>mung bean RQ</b>                 |
| <b>Oct 2022</b>            | <b>Cabomba temperature</b> +<br>photolysis         | toad caffeine uptake                                            | arsenic — <b>r_s = 0.929</b>                    | <b>polychaete</b><br>habituation    |
| <b>Jan 2023</b>            | <b>Lemna</b> nitrate                               | dilution plating, <b>k = 1.99</b>                               | Nereis — <b>t = 1.37</b><br>+ opsin             | <b>grasshoppers</b><br>field study  |
| <b>June 2023</b>           | <b>mitotic index</b> +<br>cellulose                | cactus — <b>odds ratio 19</b>                                   | pyrethrum — <b>t = 4.11</b>                     | <b>Piper betle</b><br>antimicrobial |
| <b>Oct 2023</b>            | camu camu <b>vitamin C</b><br>+ collagen           | <b>spirometer</b> O <sub>2</sub> uptake<br>+ ventilation centre | locusts — <b>t = 4.32</b>                       | <b>Cabomba</b>                      |
| <b>Jan 2024</b>            | <b>Tussilago</b><br>antibacterial +<br>prescribing | <b>pollen tube</b> 20.0 $\mu\text{m h}^{-1}$                    | lizards — <b>t = 1.96</b>                       | <b>watermelon RQ</b>                |

| Series                                 | Q1                                                                           | Q2                                                                 | Q3                                                 | Q4 (plan)                         |
|----------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------|-----------------------------------|
| <b>June 2024</b>                       | yeast <b>redox indicator</b> + NAD/FAD                                       | <b>habituation</b> — woodlice, zebrafish 71%                       | topshells — $\chi^2 = 9.3$                         | <b>mung vs pea</b> respiration    |
| <b>Oct 2024</b>                        | <b>gibberellin / starch agar</b>                                             | kenaf 10.4% + sclerenchyma                                         | pea mitotic index — <b>t = 11.3</b>                | <b>Elodea depth</b>               |
| <b>Jan 2025</b>                        | <b>brine shrimp</b> copper + inhibition                                      | acacia — $\chi^2 = 6.03$ + nitrate                                 | Aplysia — <b>r_s = 0.1</b>                         | <b>blowfly larvae</b>             |
| <b>June 2025</b>                       | <b>redox indicator</b> + anaerobic ATP                                       | earthworms: magnification, random grid, <b>SD 2.2</b>              | crude oil — <b>r_s = -0.857</b> + dilution plating | <b>fitness / spirometer</b>       |
| <b>Oct 2025</b>                        | maggot growth, 0.33 mm h <sup>-1</sup>                                       | fighting fish <b>habituation</b>                                   | cows — <b>t = 2.53</b>                             | <b>wheat</b> amylase + maltose    |
| <b>Jan 2026</b>                        | marram <b>transect</b> + xylem / phloem                                      | yeast <b>turbidity</b> 1.6 a.u. h <sup>-1</sup>                    | dung beetles — <b>t = 3.50</b>                     | <b>drumstick</b> seedlings        |
| <b>June 2026 (01)</b> ( <i>no MS</i> ) | flour beetles: length, optimum temperature, resistance                       | forest <b>diversity index D = 3.42</b>                             | training — <b>t = 0.60</b> + lactate               | <b>ampicillin vs erythromycin</b> |
| <b>June 2026 (01A)</b>                 | <b>pondweed temperature</b> , controls explained, <b>Q<sub>10</sub></b> (14) | <b>spirometer</b> + height vs tidal volume <b>r_s = 0.714</b> (17) | <b>anaerobic yeast</b> plan                        | —                                 |

### What this map shows

- **Respiration** (respirometer, redox indicator, RQ, yeast) — **ten** of the twenty papers, as a method question, a plan, or both.
- **Photosynthesis with pondweed** — **six** series.
- **Microbiology** — **eight** series (antimicrobial plans, dilution plating, turbidity, zones of inhibition).
- **Habituation** — **five** series (two plans, three method / data questions).
- **Spirometer** — **three** series + one plan.
- **The t-test data question** — **twelve** series.

## 43. What an A\* requires, and the final plan

### The rule

**A\* requires grade A for the IAL overall** ( $\geq 480$  of 600 uniform marks) **and  $\geq 90\%$  of the uniform marks across the IA2 units combined** ( $\geq 270$  of 300).

The IA2 units are **4, 5 and 6** — 120 + 120 + 60 UMS. Unit 6 is **60 of the 300**, but it is the **most predictable** of the three: the same four question types, the same mark points, the same rulings, every series. A near-full mark on Unit 6 gives you slack on Units 4 and 5.

### Priority order

| Priority | Content                                                                                     | Marks / paper |
|----------|---------------------------------------------------------------------------------------------|---------------|
| 1        | <b>Planning question</b> — preliminary, method, record/present/analyse, limitations (§1–§7) | <b>15–22</b>  |
| 2        | <b>Null hypothesis, table, graph</b> (§26–§28)                                              | <b>7</b>      |
| 3        | <b>t-test</b> calculation + conclusion (§29, §34)                                           | <b>4–5</b>    |
| 4        | <b>Describe-an-experiment template</b> and the core practicals (§8–§21)                     | <b>5–6</b>    |
| 5        | <b>Variables + control + directional effect</b> (§22)                                       | <b>3–5</b>    |
| 6        | <b>Theory chains</b> (§38)                                                                  | <b>5–12</b>   |
| 7        | Spearman, $\chi^2$ , Mann–Whitney, other calculations (§30–§33)                             | <b>3–5</b>    |
| 8        | Validity / improvements / extensions (§36–§37)                                              | <b>2–4</b>    |
| 9        | Risks and ethics (§23–§24)                                                                  | <b>1–2</b>    |

### Five-step routine for every question

1. Circle the **COMMAND WORD** and the **MARK ALLOCATION** – how many points?
2. Underline the **IV**, the **DV** and the **ORGANISM** – every method point must use these words.
3. Method points: **VALUES** (five, with units) · **APPARATUS** · **HOW** each variable is controlled · **TIME** · **REPEATS** + **MEAN** · **RATE** = ...  $\div$  time.
4. Calculations: working  $\rightarrow$  answer  $\rightarrow$  **UNIT**  $\rightarrow$  requested s.f./d.p.
5. Data: **H0** matches the test · table without an extra column · bar graph from zero · compare calculated vs critical · decide · in context · comment on range bars.

## Exam-day tactics

- **Pace:** 1.6 minutes per mark — about **25–30 minutes for the planning question**.
- **Plan in numbered sentences.** One method point per line. Examiners tick lines.
- **Say how, not just what,** for every controlled variable.
- **Never write "amount".**
- **Draw tables and graphs with a ruler and a sharp pencil;** label axes with **quantity / unit**.
- **Write the critical value and the comparison** before the decision, every time.
- **Leave five minutes** to check: every variable has a **method**, every effect is **directional**, every calculation has a **unit and precision**, every hypothesis names **both groups**.

**The Unit 6 A\* formula:** learn the **mark-point lists** in this note (they repeat almost word for word), practise **five full planning answers** against §6 under timed conditions, and do **every t-test in §29 from the raw data** until the whole data question takes under fifteen minutes.